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[Intervention Review]

Topical cyclosporine A therapy for dry eye disease

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ABSTRACT

Rationale

Ocular surface inflammation is a key component of dry eye disease (DED). Topical cyclosporine A (CsA) is commonly used to treat DED. However, uncertainty remains regarding its benefits and harms, despite extensive clinical evaluation. This updates a Cochrane review published in 2019.

Objectives

To assess the benefits and harms of topical CsA in the treatment of dry eye disease.

Search methods

We used CENTRAL, MEDLINE, Embase, two other databases, and two trial registers, together with reference checking, to identify studies included in the review. The latest search date was 17 March 2026.

Eligibility criteria

We included randomized controlled trials (RCTs) in participants with DED, regardless of age, sex, disease severity, or etiology, that compared topical CsA with its vehicle (the formulation without the active ingredient), artificial tears, or alternative CsA formulations or concentrations, with or without artificial tears.

Outcomes

We assessed: DED symptoms, ocular surface staining, tear production, tear film stability, artificial tear use, conjunctival goblet cell density, and discontinuation rates due to adverse events at 3 months.

Risk of bias

We used RoB 1 (Cochrane's original risk of bias assessment tool) to assess bias in the RCTs.

Synthesis methods

We synthesized results for each outcome using meta-analysis where possible, calculating mean differences (MDs), standardized mean differences (SMDs), or risk ratios (RRs) with 95% confidence intervals (CIs), using fixed-effect or random-effects models as appropriate. Where meta-analysis was not possible, we summarized results descriptively. We assessed the certainty of evidence using GRADE.

Included studies

We included 58 RCTs involving 10,225 participants with DED. Three countries – South Korea, the USA, and China – contributed 35 of the 58 included studies (60%). The follow-up period was typically three months (range: three days to 12 months), and the median participant age was 53 years. The proportion of women ranged from 46% to 100% (median 80%). Most studies compared topical CsA 0.05% or 0.1% with vehicle or artificial tears, while 14 studies compared different formulations or concentrations.

Synthesis of results

We grouped studies into six comparisons. Overall, the certainty of evidence was very low to moderate, downgraded for imprecision, risk of bias, or heterogeneity.

CsA 0.05% versus vehicle, artificial tears, or both. CsA 0.05% may improve DED symptoms at three months compared with vehicle, artificial tears, or both (SMD -0.33 , 95% CI -0.65 to -0.01 ; $I^2 = 86\%$; 13 studies, 1396 participants; low-certainty evidence). CsA 0.05% may also improve corneal staining (SMD -0.20 , 95% CI -0.31 to -0.09 ; $I^2 = 0\%$; 7 studies, 1355 participants; low-certainty evidence), while the effect estimate was imprecise for conjunctival staining (SMD -0.21 , 95% CI -0.52 to 0.09 ; $I^2 = 65\%$; 3 studies, 582 participants; low-certainty evidence). Participants treated with CsA 0.05% were probably more likely to discontinue treatment due to adverse events (RR 1.99, 95% CI 1.02 to 3.87; $I^2 = 23\%$; 9 studies, 2109 participants; moderate-certainty evidence). For other outcomes, low- or very low-certainty evidence favored CsA 0.05% over vehicle or artificial tears.

Other CsA formulations or concentrations versus CsA 0.05%. Fourteen studies compared different CsA formulations (e.g. nanoemulsion or cationic emulsion) or concentrations (e.g. 0.1%) with CsA 0.05%. Effect estimates were imprecise for all outcomes, except tear production, which favored other CsA formulations in the pooled analysis (MD 0.30, 95% CI 0.17 to 0.43; $I^2 = 0\%$; 5 studies, 572 participants), and CsA 0.08% nanoemulsion in the formulation subgroup analysis (MD 0.30, 95% CI 0.17 to 0.43; 1 study, 155 participants), both with low-certainty evidence. CsA 0.05% likely had a lower rate of discontinuation due to adverse events than other CsA formulations (RR 1.72, 95% CI 1.07 to 2.78; $I^2 = 0\%$; 6 studies, 1421 participants; moderate-certainty evidence).

CsA 0.1% versus vehicle, artificial tears, or both. The evidence was mixed for DED symptoms and imprecise for tear stability (MD 0.15, 95% CI -0.11 to 0.42; $I^2 = 0\%$; 3 studies, 979 participants; low-certainty evidence). CsA 0.1% probably improves corneal staining (SMD -0.16 , 95% CI -0.24 to -0.08 ; $I^2 = 15\%$; 7 studies, 2940 participants) and conjunctival staining (SMD -0.20 , 95% CI -0.28 to -0.12 ; $I^2 = 0\%$; 5 studies, 2303 participants), both with moderate-certainty evidence, and may improve tear production (MD 0.66, 95% CI 0.09 to 1.24; $I^2 = 30\%$; 3 studies, 735 participants; low-certainty evidence). Participants treated with CsA 0.1% were probably more likely to discontinue due to adverse events (RR 1.58, 95% CI 1.04 to 2.41; $I^2 = 0\%$; 7 studies, 3051 participants; moderate-certainty evidence).

CsA 0.1% versus CsA 0.1% plus artificial tears. One study evaluating 0.1% CsA, with or without artificial tears, in moderate-to-severe DED found very low-certainty evidence of no significant difference in the assessed outcomes.

CsA 1% versus vehicle. One study (60 participants) indicated that CsA 1% may improve DED symptoms (MD -2.34 , 95% CI -3.70 to -0.98), corneal staining (MD -0.33 , 95% CI -0.56 to -0.10), tear production (MD 2.01, 95% CI 1.11 to 2.91), and tear stability (MD 2.77, 95% CI 2.14 to 3.40), but the evidence is very uncertain. No participants discontinued due to adverse events.

CsA 2% versus vehicle. Two studies compared these treatments. CsA 2% may improve DED symptoms and ocular staining (MD -1.82 , 95% CI -2.19 to -1.45 ; 1 study, 30 participants), and may have little to no effect on tear production (MD 0.35, 95% CI -0.24 to 0.94; 1 study, 30 participants) or tear film stability (MD 2.82, 95% CI 2.17 to 3.47; 1 study, 30 participants), but the evidence is very uncertain. Discontinuation rates were minimal.

Authors' conclusions

Low-certainty evidence suggests that topical CsA may improve DED symptoms and certain clinical signs compared with vehicle or artificial tears. However, results for DED symptoms and tear parameters were inconsistent and highly heterogeneous. Higher CsA concentrations were associated with increased treatment discontinuation. Future studies should stratify DED by subtype and severity, compare formulations, and assess long-term benefit and harm.

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Registration

Original review (2019) DOI: 10.1002/14651858.CD010051.pub2

PLAIN LANGUAGE SUMMARY

What are the benefits and risks of cyclosporine A (CsA) eye drops for treating dry eye disease?

Key messages

- Cyclosporine A (CsA) eye drops, including Restasis, Cequa, and other newer versions, may improve some signs and symptoms of dry eye compared with their "vehicle" (inactive eye drops containing no CsA) or artificial tears. However, we have only moderate or low confidence in these results.
- People who used CsA eye drops were slightly more likely to stop treatment due to unwanted effects, but overall, few people withdrew from studies for this reason.
- We need higher-quality, longer-lasting studies that look at the benefits and harms of CsA eye drops in people with dry eye disease of varying seriousness.

What is dry eye disease?

The eye's surface is moist and covered with a thin film of tears. When problems arise with the tear film, people can develop a common condition known as dry eye disease. People with dry eye may have eye discomfort (such as a feeling of grittiness), burning sensations, light sensitivity, and blurred or fluctuating vision, which can affect driving and reading and lower their quality of life.

Some people develop dry eye disease because of another health condition or as a result of eye treatment, while in others, the cause is unknown. Many cases of dry eye are linked to health conditions that cause inflammation (swelling and irritation) in and around the eyes.

What are cyclosporine A (CsA) eye drops?

CsA eye drops are anti-inflammatory medicines that aim to reduce eye surface inflammation, improve tear production, and prevent eye surface damage. CsA eye drops come in different strengths, typically ranging from 0.05% (lower strength) to 2% (higher strength).

What did we want to find out?

We wanted to find out whether CsA eye drops work better than eye drops containing all the same ingredients except cyclosporine ('vehicle eye drops' or just 'vehicle'), artificial tears, or another type or strength of CsA to improve:

- dry eye symptoms;
- eye surface health;
- tear production;
- how well tears stay on the eye (tear film stability).

We also wanted to find out whether people using CsA eye drops were more likely to stop using them because of unwanted effects than those using vehicle, artificial tears, or CsA drops of a different type or strength.

What did we do?

We searched for studies comparing CsA eye drops with vehicle, artificial tears, or another CsA formulation or strength. We compared and summarized the results of these studies and rated our confidence in the evidence based on factors such as study methods and the number of people in the studies.

What did we find?

We found 58 studies involving 10,225 people with dry eye disease. China, South Korea, and the USA contributed 35 of the 58 studies (60%). Participants were typically around 53 years old. Women made up about 80% of participants overall, though proportions varied across studies from 46% to 100%. Pharmaceutical companies funded or sponsored nearly half of the studies. Non-profit organizations supported 10 studies, six studies had no external funding, and 14 studies did not report funding sources.

We grouped the studies into six groups based on CsA strength and the comparison treatments. Here, we focus on three groups with the most data.

Main results

Comparison 1

Compared to vehicle, artificial tears, or both, using CsA 0.05% eye drops for three months may have slightly improved people's:

- dry eye symptoms (13 studies, 1396 people);
- eye surface health (7 studies, 1355 people).

Results for tear production and tear film stability were inconsistent, preventing us from drawing firm conclusions.

Comparison 2

People using CsA eye drops of different strengths and types may have had slightly better tear production than those using CsA 0.05% eye drops after three months, but the difference between groups was small (5 studies, 572 people). We are uncertain about the effects on:

- dry eye symptoms;
- eye surface health;
- tear film stability.

Comparison 3

Compared to vehicle, artificial tears, or both, using CsA 0.1% eye drops for three months:

- probably improved eye surface health slightly (7 studies, 2940 people);
- may have improved tear production slightly (3 studies, 735 people).

Results for dry eye symptoms were inconsistent, preventing us from drawing firm conclusions. We are uncertain about the effect of CsA 0.1% on tear film stability.

Stopping treatment due to unwanted effects

Compared with people using vehicle eye drops or artificial tears, those using CsA 0.05% or 0.1% were slightly more likely to stop treatment due to unwanted effects. Overall, only a few people stopped treatment because of unwanted effects.

What are the limitations of the evidence?

We have little confidence in the evidence for dry eye symptoms, eye surface health, tear production, and tear film stability because study results were often inconsistent or too uncertain to determine the true size of the effect. Many studies gave unclear or few details about their methods. However, we are moderately confident in the evidence about stopping treatment.

How up to date is this evidence?

The evidence is current to March 2026.

SUMMARY OF FINDINGS

Summary of findings 1. Topical cyclosporine A 0.05% versus vehicle or artificial tears for dry eye disease

Topical cyclosporine A 0.05% versus vehicle or artificial tears for dry eye disease

Population: people with dry eye disease

Intervention: topical CsA 0.05%

Comparison: vehicle plus artificial tears (AT), or artificial tears alone

Setting: outpatient

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	Number of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Assumed risk with vehicle or artificial tears	Corresponding risk with CsA 0.05%				
Change from baseline in subjective symptoms of dry eye, assessed with OSDI or SANDE (0-to-100 scale; lower scores indicates better condition) at 3 months	Mean change ranged from -17.04 to -12.11, or mean ranged from 14 to 44.6	SMD in the intervention groups was on average 0.33 smaller (small-to-moderate effect in favor of CsA) (95% CI -0.65 to -0.01)	-	1396 (13)	⊕⊕⊕⊕ Low ^{a,b}	Direction of effect among the studies not included in the meta-analysis: 9 studies favored the intervention and 3 favored control.
Change from baseline in ocular surface dye staining (lower scores indicate better condition) at 3 months	Corneal staining: mean change from baseline SMD** 0.20 smaller (small effect in favor of CsA) (95% CI -0.31 to -0.09)			1355 (7)	⊕⊕⊕⊕ Low ^{a,b}	Direction of effect among the studies not included in the meta-analysis: 5 studies favored CsA 0.05% while 2 studies favored control.
	Corneal staining: mean No pooled estimate due to considerable heterogeneity and inconsistency in direction of effect (9 studies favored CsA and 2 favored control)			1278 (11)		
	Conjunctival staining: mean change from baseline			582 (3)		Direction of effect among the studies not included in the meta-analysis:

	SMD** 0.21 smaller (small effect in favor of CsA) (95% CI -0.52 to 0.09)				sis: 2 studies favored CsA 0.05% while 2 studies favored control.	
	Conjunctival staining: mean No pooled estimate due to considerable heterogeneity (2 studies favored CsA)			106 (2)		
Change from baseline in aqueous tear production, assessed with Schirmer test (mm/5 minutes; higher values indicate better condition) at 3 months	No pooled estimate due to considerable heterogeneity (11 studies favored CsA and 7 studies favored control)			1487 (18)	⊕⊕○○ Low ^{a,b}	Direction of effect among the studies not included in the meta-analysis: 8 studies favored CsA 0.05% while 1 study favored artificial tears.
Change from baseline in tear film stability, assessed as tear film break-up time (seconds; higher values indicate better condition) at 3 months	No pooled estimate due to considerable heterogeneity (17 studies favored CsA and 3 studies favored control)			2777 (20)	⊕⊕○○ Low ^{a,b}	Direction of effect among the studies not included in the meta-analysis: 7 studies favored CsA 0.05% and 1 showed no difference.
Change from baseline in frequency of artificial tear usage (lower values indicate better condition) at 3 months	See comment				⊕⊕○○ Very Low ^{a-c}	No between-group difference in adjunctive usage of AT in 2 studies. People in the CsA 0.05% group used AT less frequently than those in the vehicle group (P = 0.006) in 1 study.
Change from baseline in conjunctival goblet cell density, assessed as cells per unit (higher values indicate better condition) at 6 months	No pooled estimate due to considerable heterogeneity (all 4 studies favored CsA)			248 (4)	⊕⊕○○ Low ^{a,b}	We were unable to include 4 trials in the meta-analysis due to lack of data or insufficient reporting.
Proportion of participants who discontinued study due to adverse events (lower is better) during the study period	23 per 1000	46 per 1000 (23 to 178)	RR 1.99 (1.02 to 3.87)	2109 (9)	⊕⊕○○ Moderate ^a	–

*The risk in the intervention group (and its 95% CI) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

**0.2 represents a small effect, 0.5 represents a moderate effect, and 0.8 represents a large effect according to Cohen's effect sizes.

AT: artificial tears; **CI:** confidence interval; **CsA:** cyclosporine A; **OSDI:** Ocular Surface Disease Index; **RR:** risk ratio; **SANDE:** Symptom Assessment in Dry Eye; **SMD:** standardized mean difference

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

^aDowngraded by one level for risk of bias

^bDowngraded by one level for unexplained heterogeneity or inconsistency

^cDowngraded by one level for imprecision of results due to crossing line of no effect

Summary of findings 2. Other CsA formulations or concentrations for dry eye disease versus topical cyclosporine A 0.05%

Other CsA formulations and concentrations versus CsA 0.05% for dry eye disease

Population: people with dry eye disease

Intervention: other formulations or concentrations of CsA

Comparison: topical CsA 0.05%

Setting: outpatient

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	Number of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Assumed risk with CsA 0.05%	Corresponding risk with other CsA formulations				
Change from baseline in subjective symptoms of dry eye, assessed with OSDI, SANDE, or VAS (0-to-100 scale; lower scores indicate better condition) at 3 months	Mean change ranged from -32.07 to -6.74, or mean 36.82	Mean change in intervention groups was on average 0.20 greater (95% CI -2.90 to 3.30)	-	750 (7)	⊕⊕⊕⊕ Low ^{a,b}	Subgroup analysis based on formulation: CsA 0.1% cationic emulsion (MD 6.63, 95% CI -7.96 to 21.22; 1 study, 44 participants), CsA 0.05% nanoemulsion (MD 1.27, 95% CI -3.63 to 6.17; I ² = 0%; 3 studies, 225 participants), CsA 0.08% nanoemulsion (MD 0.42, 95% CI -5.30 to 6.14; 1 study, 155 participants), CsA 0.05% nanoparticulate formulation (MD -3.80, 95% CI -14.40 to 6.80; 1 study, 95 participants), and gel-based formulation (SMD -0.10, 95% CI -0.40 to 0.20; 1 study, 232 participants). Direction of effect among the studies not included in the meta-analysis: 1 study favored CsA 0.1%



						and another study showed no difference between groups (CsA 0.1% versus Restasis).
Change from baseline in ocular surface dye staining (lower scores indicate better condition) at 3 months	Corneal staining: mean change from baseline			572	⊕⊕⊕⊕ Low ^{a,c}	Subgroup analysis based on formulation: CsA 0.05% nanoemulsion (SMD -0.46, 95% CI -1.15 to 0.23; $I^2 = 83\%$; 3 studies, 225 participants), CsA 0.08% nanoemulsion (SMD -0.06, 95% CI -0.38 to 0.25; 1 study, 155 participants), and CsA 0.05% nanoparticulate formulation (SMD -0.23, 95% CI -0.51 to 0.05; 1 study, 95 participants).
	SMD** 0.24 smaller (small effect in favor of other formulation of CsA)			(5)		
	(95% CI -0.49 to 0.00)					
	Corneal staining: mean 1.31	Corneal staining: mean on average 0.08 greater (-0.78 to 0.94)	-	44 (1)		Direction of effect among the studies not included in the meta-analysis: 1 study favored CsA 0.05% nanoemulsion, 1 study favored CsA 0.1% compared with CsA 0.05%, and 3 studies showed no evidence of a difference.
	Conjunctival staining: mean change from baseline			572	⊕⊕⊕⊕ Low ^{a,b}	Direction of effect among the studies not included in the meta-analysis: 1 study favored CsA 0.1% compared with CsA 0.05%.
	SMD** 0.17 smaller (small effect in favor of other formulation of CsA)			(5)		
	(95% CI -0.36 to 0.01)					
	Conjunctival staining: mean 1.46	Conjunctival staining: mean on average 0.11 greater (-1.27 to 1.49)	-	44 (1)		
Change from baseline in aqueous tear production, assessed with Schirmer test (mm/5 minutes; higher values indicate better condition) at 3 months	Mean change from -0.5 to 4.65	Mean change in intervention groups on average 0.30 greater (0.17 to 0.43)	-	572 (5)	⊕⊕⊕⊕ Low ^{a,b}	Subgroup analysis based on formulation: CsA 0.05% nanoemulsion (MD -0.07, 95% CI -1.23 to 1.09; $I^2 = 0\%$; 3 studies, 225 participants), CsA 0.05% nanoparticulate formulation (MD 2.30, 95% CI -0.14 to 4.74; 1 study, 192 participants), and CsA 0.08% nanoemulsion compared with Restasis (MD 0.30, 95% CI 0.17 to 0.43; 1 study, 155 participants).
						Direction of effect among the studies not included in the meta-analysis: 1 study favored CsA 0.1% compared with CsA 0.05%, and 4 studies showed no evidence of a difference.
Change from baseline in tear film stability,	Mean change from 0.36 to 1.53, mean 5.01	Mean change in intervention groups on average 0.30 greater (0.17 to 0.43)	-	424 (5)	⊕⊕⊕⊕ Low ^{a,b}	Subgroup analysis based on formulation: CsA 0.05% nanoemulsion (MD 0.62, 95% CI 0.06 to 1.18; $I^2 = 0\%$; 3 studies, 225 participants), CsA 0.1%

assessed as tear film break-up time (seconds; higher values indicate better condition) at 3 months	age 0.38 greater (−0.03 to 0.78)					cationic emulsion (MD 0.49, 95% CI −0.99 to 1.97; 1 study, 44 participants), and 0.08% nanoemulsion (MD 0.04, 95% CI −0.60 to 0.68; 1 study, 155 participants) compared with Restasis. Direction of effect among the studies not included in the meta-analysis: 3 studies showed no evidence of a difference.
Change from baseline in frequency of artificial tear usage (lower values indicate better condition) at 3 months	See comments				⊕⊕⊕⊕ Low ^{a,b}	Sum of artificial tear use during 8 weeks of trial comparing CsA 0.08% nanoemulsion group and Restasis: MD 56.10, 95% CI −5.30 to 117.50; 1 study, 155 participants
Change from baseline in conjunctival goblet cell density, assessed as cells per unit (higher values indicate better condition) at 6 months	Mean change −0.39	Mean change in the intervention group was 0.31 greater (−0.51 to 1.13)	−	36 (1)	⊕⊕⊕⊕ Low ^{a,b}	−
Proportion of participants who discontinued study due to adverse events (lower is better) during the study period	35 per 1000	60 per 1000 (37 to 97)	RR 1.72 (1.07 to 2.78)	1421 (6)	⊕⊕⊕⊕ Moderate ^a	−

***The risk in the intervention group** (and its 95% CI) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

**0.2 represents a small effect, 0.5 represents a moderate effect, and 0.8 represents a large effect according to Cohen's effect sizes.

AT: artificial tears; **CI:** confidence interval; **CsA:** cyclosporine A; **MD:** mean difference; **OSDI:** Ocular Surface Disease Index; **RR:** risk ratio; **SANDE:** Symptom Assessment in Dry Eye; **SMD:** standardized mean difference; **VAS:** visual analog scale

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

^aDowngraded by one level for risk of bias
^bDowngraded by one level for unexplained heterogeneity or inconsistency of results
^cDowngraded by one level for imprecision of results due to crossing line of no effect

Summary of findings 3. Topical cyclosporine A 0.1% versus vehicle or artificial tears for dry eye disease

Topical cyclosporine A 0.1% versus vehicle or artificial tears for dry eye disease

Population: people with dry eye disease

Intervention: topical CsA 0.1%

Comparison: vehicle plus artificial tears, or artificial tears alone

Setting: outpatient

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	Number of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Assumed risk with vehicle or artificial tears	Corresponding risk with CsA 0.1%				
Change from baseline in subjective symptoms of dry eye, assessed with OSDI, SANDE, or VAS (0-to-100 scale; lower scores indicate better condition) at 3 months	No pooled estimate due to considerable heterogeneity (5 studies favored CsA and 4 studies favored control)			3217 (9)	⊕⊕⊕⊕ Low ^{a,b}	Direction of effect among the studies not included in the meta-analysis: 1 study favored CsA 0.1% and 1 study showed no evidence of a difference.
Change from baseline in ocular surface dye staining (lower scores indicate better condition) at 3 months	Corneal staining: mean change from baseline SMD** 0.16 smaller (small effect in favor of CsA) (95% CI -0.24 to -0.08)			2940 (7)	⊕⊕⊕⊕ Moderate ^a	Direction of effect among the studies not included in the meta-analysis: 1 study favored CsA 0.1% and 1 study showed no evidence of a difference.
	Conjunctival staining: mean change from baseline SMD** 0.20 smaller (small effect in favor of CsA) (95% CI -0.28 to -0.12)			2303 (5)		Direction of effect among the studies not included in the meta-analysis: 1 study favored CsA 0.1%.
Change from baseline in aqueous tear production, assessed with Schirmer test	Mean change from baseline -0.08 to 1.5	Mean change in intervention groups on aver-	-	735 (3)	⊕⊕⊕⊕ Low ^{a,c}	Direction of effect among the studies not included in the meta-analysis: 1 study

(mm/5 minutes; higher values indicate better condition) at 3 months		age 0.66 greater (0.09 to 1.24)					favored CsA 0.1% and 2 studies showed no evidence of a difference.
Change from baseline in tear film stability, assessed as tear film break-up time (seconds; higher values indicate better condition) at 3 months	Mean change from baseline 0.33 to 1.13	Mean change in the intervention groups on average 0.15 greater (−0.11 to 0.42)	–	979 (3)	⊕⊕⊕⊕ Low ^{a,c}	–	
Change from baseline in frequency of artificial tear usage (lower values indicate better condition) at 3 months	No studies examined this outcome					–	
Change from baseline in conjunctival goblet cell density, assessed as cells per unit (higher values indicate better condition) at 6 months	No studies examined this outcome					–	
Proportion of participants who discontinued study due to adverse events (lower is better) during the study period	23 per 1000	36 per 1000 (24 to 55)	RR 1.58 (1.04 to 2.41)	3051 (7)	⊕⊕⊕⊕ Moderate ^a	–	

***The risk in the intervention group** (and its 95% CI) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

**0.2 represents a small effect, 0.5 represents a moderate effect and 0.8 represents a large effect according to Cohen's effect sizes.

AT: artificial tears; **CI:** confidence interval; **CsA:** cyclosporine A; **MD:** mean difference; **OSDI:** Ocular Surface Disease Index; **RR:** risk ratio; **SANDE:** Symptom Assessment in Dry Eye; **SMD:** standardized mean difference; **VAS:** visual analog scale

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

^aDowngraded by one level for risk of bias

^bDowngraded by one level for unexplained heterogeneity or inconsistency of results

^cDowngraded by one level for imprecision of results due to crossing line of no effect

Summary of findings 4. Topical cyclosporine A 0.1% versus topical CsA 0.1% plus artificial tears for dry eye disease

Topical cyclosporine A 0.1% versus topical CsA 0.1% + artificial tears for dry eye disease

Population: people with dry eye disease

Intervention: topical CsA 0.1% (Ikervis)

Comparison: topical CsA 0.1% (Ikervis) plus artificial tears

Setting: outpatient

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	Number of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Assumed risk with CsA 0.1% + artificial tears	Corresponding risk with CsA 0.1%				
Change from baseline in subjective symptoms of dry eye, assessed with OSDI (0-to-100 scale; lower values indicate better condition) at 1 month	Mean at baseline 8.4	Mean in the intervention group 1.02 greater (-2.26 to 4.30)	–	19 (1)	⊕○○○ Very low ^{a,b}	–
Change from baseline in ocular surface dye staining (lower values indicate better condition) at 1 month	See comment			19 (1)	⊕○○○ Very low ^{a,b}	"Distribution of participants with degree of ocular staining did not show the difference between the intervention groups after one month of treatment (P=0.828)"
Change from baseline in aqueous tear production, assessed with Schirmer test (mm/5 minutes; higher values indicate better condition) at 1 month	Mean at baseline 10	Mean in the intervention group 0.33 greater (-5.39 to 6.05)	–	19 (1)	⊕○○○ Very low ^{a,b}	–
Change from baseline in tear film stability, assessed as tear film break-up time (seconds; higher values indicate better condition) at 1 month	Mean at baseline 8.11	Mean in the intervention group 0.55	–	19 (1)	⊕○○○ Very low ^{a,b}	–

	greater (-2.89 to 1.79)	
Change from baseline in frequency of artificial tear usage (lower values indicate better condition) at 3 months	This study did not examine this outcome.	–
Change from baseline in conjunctival goblet cell density, assessed as cells per unit (higher values indicate better condition) at 6 months	This study did not examine this outcome.	–
Proportion of participants who discontinued study due to adverse events (lower is better) during the study period	This study did not examine this outcome.	–

***The risk in the intervention group** (and its 95% CI) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

AT: artificial tears; **CI:** confidence interval; **CsA:** cyclosporine A; **MD:** mean difference; **OSDI:** Ocular Surface Disease Index; **SMD:** standardized mean difference

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

^aDowngraded by one level for indirectness

^bDowngraded by two levels for imprecision of results due to crossing line of no effect and small sample size

Summary of findings 5. Topical cyclosporine A 1% versus vehicle for dry eye disease

Topical cyclosporine A 1% versus vehicle for dry eye disease

Population: people with dry eye disease

Intervention: topical CsA 1% plus artificial tears

Comparison: vehicle plus artificial tears

Setting: outpatient

Outcomes	Anticipated absolute effects* (95% CI)	Relative effect (95% CI)	Number of participants (studies)	Certainty of the evidence (GRADE)	Comments
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	Assumed risk with vehicle	Corresponding risk with CsA 1%				
Change from baseline in subjective symptoms of dry eye (0-to-15 scale; lower score indicates better condition) at 3 months	Mean 11.57	Mean in the intervention group 2.34 smaller (−3.70 to −0.98)	–	60 (1)	⊕⊕⊕⊕ Very low ^{a,b}	Direction of effect among the studies not included in the meta-analysis: 1 study favored CsA 1%.
Change from baseline in ocular surface dye staining (lower scores indicate better condition) at 3 months	Mean 1.92	Mean in the intervention group 0.33 smaller (−0.56 to −0.10)	–	60 (1)	⊕⊕⊕⊕ Very low ^{a,b}	Direction of effect among the studies not included in the meta-analysis: 1 study favored CsA 1%.
Change from baseline in aqueous tear production, assessed with Schirmer test (mm/5 minutes; higher scores indicate better condition) at 3 months	Mean 8.44	Mean in the intervention group 2.01 greater (1.11 to 2.91)	–	60 (1)	⊕⊕⊕⊕ Very low ^{a,b}	Direction of effect among the studies not included in the meta-analysis: 1 study favored CsA 1%.
Change from baseline in tear film stability, assessed as tear film break-up time (seconds; higher values indicate better condition) at 3 months	Mean 5.92	Mean in the intervention group 2.77 greater (2.14 to 3.40)	–	60 (1)	⊕⊕⊕⊕ Very low ^{a,b}	Direction of effect among the studies not included in the meta-analysis: 1 study favored CsA 1%.
Change from baseline in frequency of artificial tear usage (lower values indicate better condition) at 3 months	No studies examined this outcome.					–
Change from baseline in conjunctival goblet cell density, assessed as cells per unit (higher values indicate better condition) at 6 months	No studies examined this outcome.					–
Proportion of participants who discontinued study due to adverse events (lower is better) during the study period	See comment				⊕⊕⊕⊕ Very low ^{a,b}	1 study reported that none of the participants discontinued the study due to adverse events.

^aThe risk in the intervention group (and its 95% CI) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CsA: cyclosporine A; **CI:** confidence interval

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

^aDowngraded by one level for risk of bias

^bDowngraded by two levels for imprecision of results due to crossing line of no effect and small sample size

Summary of findings 6. Topical cyclosporine A 2% versus vehicle for dry eye disease

Topical cyclosporine A 2% versus vehicle for dry eye disease

Population: people with dry eye disease

Intervention: topical CsA 2% plus artificial tears

Comparison: vehicle plus artificial tears

Setting: outpatient

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	Number of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Assumed risk with vehicle	Corresponding risk with CsA 2%				
Change from baseline in subjective symptoms of dry eye at 3 months	See comments				⊕⊕⊕⊕ Very low ^{a,b}	Direction of effect: 1 study favored CsA 2%
Change from baseline in ocular surface dye staining (lower scores indicate better condition) at 3 months	Mean at base-line 5.2	Mean in the intervention group 1.82 smaller (−2.19 to −1.45)	–	30 (1)	⊕⊕⊕⊕ Very low ^{a,b}	Direction of effect of another study: no difference
Change from baseline in aqueous tear production, assessed with Schirmer test (mm/5 min-	Mean at base-line 5.5	Mean in the intervention group	–	30 (1)	⊕⊕⊕⊕ Very low ^{a,b}	Direction of effect of another study: no difference



utes; higher values indicate better condition) at 3 months		0.35 greater (-0.24 to 0.94)				
Change from baseline in tear film stability, assessed as tear film break-up time (seconds; higher values indicate better condition) at 3 months	Mean at baseline 5.7	Mean in the intervention group 2.82 greater (2.17 to 3.47)	-	30 (1)	⊕⊕⊕⊕ Very low ^{a,b}	Direction of effect of another study: no difference
Change from baseline in frequency of artificial tear usage (lower values indicate better condition) at 3 months	No studies examined this outcome.					-
Change from baseline in conjunctival goblet cell density, assessed as cells per unit (higher values indicate better condition) at 6 months	No studies examined this outcome.					-
Proportion of participants who discontinued study due to adverse events (lower is better) during the study period	See comments				⊕⊕⊕⊕ Very low ^{a,b}	1 study reported that none of the participants discontinued the treatment due to adverse events.

*The risk in the intervention group (and its 95% CI) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CsA: cyclosporine A; CI: confidence interval

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

^aDowngraded by one level for risk of bias

^bDowngraded by two levels for imprecision of results due to crossing line of no effect and small sample size

BACKGROUND

Description of the condition

Dry eye disease (DED), commonly defined as tear dysfunction, occurs when the lacrimal functional unit (comprised of the lacrimal glands, ocular surface, and eyelids, and the sensory and motor nerves connecting them) is no longer able to maintain a stable precorneal tear layer. The pathophysiology of DED is complex and involves several pathways, including inflammatory pathways. DED may develop from dysfunction or disease of one or more components of the lacrimal functional unit due to aging, inflammation, hormonal changes, secondary to refractive or cataract surgery, secondary to a local ocular disease such as glaucoma, or systemic autoimmune diseases such as Sjögren's disease, lupus, rheumatoid arthritis, or diabetes.

Epidemiology

DED is a common multifactorial ocular surface disease and an increasingly important public health concern worldwide. It can substantially impair visual function and quality of life and may become chronic or progressive if inadequately managed. Recent estimates suggest that approximately 9% to 12% of the global population is affected [1], although prevalence varies considerably across populations and regions. For example, rate estimates range from as low as 4% in North America to nearly 48% in Africa [1, 2]. In general, higher prevalence has been reported in parts of Africa, Asia, and the Middle East than in North America and Europe [1, 2, 3, 4]. DED is also increasingly recognized in children and adolescents, with recent studies suggesting that the rising prevalence is associated with factors such as increased digital device use and the wearing of contact lenses [5, 6].

Accurate estimation of DED prevalence remains challenging because diagnostic criteria vary widely among studies. Some studies rely primarily on symptoms, whereas others require both symptoms and objective clinical signs. Differences in case definitions, study populations, and diagnostic methods contribute substantially to heterogeneity in reported prevalence estimates and make it difficult to determine the prevalence of specific DED subtypes (aqueous-deficient, evaporative, or mixed) and disease severity categories. Studies applying the Tear Film & Ocular Surface Society (TFOS) Dry Eye Workshop (DEWS) II criteria, which incorporate both DED signs and symptoms, have reported DED prevalence rates of approximately 29% to 44% [1, 4, 7].

A further challenge in DED is the frequent mismatch between clinical signs and patient-reported symptoms, together with the lack of a single diagnostic gold standard. Some people report bothersome symptoms despite minimal clinical signs, whereas others show clinical signs with relatively few symptoms. This discordance contributes to the variability in reported prevalence rates. Recent studies have suggested that underlying systemic diseases may influence how DED manifests. In some individuals, this results in more severe symptoms than observable clinical signs, while in others, clinical signs are more pronounced than the reported symptoms [8, 9, 10].

Among several risk factors, female sex and increasing age are consistently associated with a higher prevalence of DED, with rates rising in both men and women with each decade of life after the age of 40 [11, 12]. Across all age groups, more women are

affected than men [11, 12]. In recent years, evidence supporting various risk factors for DED has strengthened. Studies have identified a wide range of factors associated with DED, including ocular surface conditions such as meibomian gland dysfunction (MGD), systemic autoimmune connective tissue diseases including Sjögren's disease, hormonal factors such as androgen deficiency, behavioral exposures such as prolonged computer use and contact lens wear, and medical conditions including hematopoietic stem cell transplantation, metabolic, mental health, vascular, and neurological disorders, as well as malignancy [8, 10, 13, 14, 15, 16].

Description of the intervention and how it might work

Topical cyclosporine A (CsA) is an anti-inflammatory agent used to treat DED. Several proprietary ophthalmic formulations of cyclosporine A (CsA) have been developed for the treatment of dry eye disease (DED), differing in concentration, formulation, and drug delivery system. These include Restasis (0.05% ophthalmic emulsion), Cequa (0.09% ophthalmic solution), Vevye (0.1% water-free ophthalmic solution), and Ikervis (0.1% preservative-free cationic emulsion). These formulations have been approved in various jurisdictions for the treatment of DED and have been shown to improve signs of ocular surface disease, including tear production and ocular surface staining.

CsA has been widely used as an immunomodulatory therapy to prevent or control rejection of solid organ transplants and to treat autoimmune diseases, including non-infectious uveitis, by inhibiting the activation and function of T lymphocytes [17]. The immunomodulation activity of CsA is relevant to DED because activated CD4⁺ T cells have been found to infiltrate the conjunctiva (the transparent membrane covering the outer surface of the eyeball, except the cornea, and the inner surface of the eyelids) of people with both Sjögren's disease (a chronic autoimmune disease characterized by dry eye and dry mouth) and non-Sjögren's disease-associated DED. CsA, by binding to cyclophilin D, is thought to be primarily responsible for its inhibition of apoptosis. CsA dampens T cell proliferation by preventing NFAT dephosphorylation and thereby reducing IL-2 transcription and T-cell activation/proliferation, causing programmed cell death [18]. The CsA-cyclophilin D complex binds to and prevents opening of the mitochondrial permeability transition pore [19]. Opening of this pore in response to cellular stress or damage is an early step in the apoptosis cascade. Increased apoptosis in the ocular surface epithelium, particularly the conjunctiva, has also been found in dry eye. CsA significantly reduced apoptosis of conjunctival epithelial cells, as assessed by DNA fragmentation and levels of activated caspase-3, and a decrease in interferon-gamma and interleukin-17 production in an experimental murine model of dry eye [20, 21, 22, 23].

Why it is important to do this review

There is no clear consensus regarding the efficacy of different strengths and formulations of topical CsA, or whether they are superior or equivalent to artificial tears, placebo, or vehicle controls. In ophthalmic trials, a vehicle control contains the same formulation as the active treatment but without cyclosporine. Because vehicle formulations may themselves improve dry eye symptoms and signs, comparisons with vehicle controls are important for isolating the effects of CsA.

With growing evidence about the role of the immune system in DED pathophysiology, CsA has become one of the most widely used immunomodulatory treatments for chronic DED. However, DED is a multifactorial condition, and people present with varying disease subtypes, severity, and duration. Consequently, it remains uncertain whether the benefits of CsA are consistent across patient populations or formulations. Given the costs, availability constraints, and potential adverse effects associated with treatment, a critical appraisal of the evidence is important to inform clinical practice and future research.

This review updates a Cochrane review first published in 2019, which included 30 studies identified up to February 2018 [24]. Since then, additional clinical trials have been published, including studies evaluating newer CsA formulations. An updated synthesis of the evidence is therefore warranted to assess the benefits and harms of topical CsA for the treatment of DED.

OBJECTIVES

To assess the benefits and harms of topical CsA in the treatment of dry eye disease.

METHODS

We followed the Methodological Expectations of Cochrane Intervention Reviews (MECIR) standards when conducting this review and PRISMA 2020 in reporting our findings [25, 26].

Because we identified a sufficient number of full-text publications in this update, we decided to exclude conference abstracts due to their limited data reporting and insufficient methodological details [27]. We revised the main assessment time point to three months, as most studies comparing CsA with vehicle, artificial tears, or different formulations and concentrations of CsA reported outcomes at 12 weeks or three months. In contrast, only one-fourth of the included studies reported outcomes at six months. Hence, we adopted the 3-month time point to ensure greater consistency and comparability across studies. We also modified the title, using the term 'dry eye disease' instead of 'dry eye syndrome'.

Criteria for considering studies for this review

Types of studies

We included only RCTs in this review.

Types of participants

We included participants with DED, defined as evidence of tear dysfunction based on DED symptoms, and the presence of an unstable tear film or ocular surface damage/disease associated with DED. We included participants of any age, sex, chronicity, severity, or classification of DED, including Sjögren's disease-associated and non-Sjögren's disease-associated DED.

Types of interventions

We considered RCTs in which topical formulations of CsA were compared with one another (head-to-head comparison) or with artificial tears, placebo, or vehicle. We also considered trials comparing CsA with artificial tears to artificial tears alone.

Outcome measures

We considered three months the primary assessment time point. For studies that did not report outcomes at three months, we extracted data from the follow-up time point closest to three months.

Critical outcomes

Our critical outcome was mean change from baseline in subjective symptoms of DED at three months' follow-up, measured by validated questionnaires, including but not limited to the Ocular Surface Disease Index (OSDI), Standard Patient Evaluation of Eye Dryness (SPEED), and Symptom Assessment in Dry Eye (SANDE) instruments [28, 29, 30].

Important outcomes

- Ocular surface dye staining, measured as the mean change in corneal fluorescein staining scores, or conjunctival lissamine green or rose Bengal staining scores, or both, at three months.
- Aqueous tear production, measured as the mean change in Schirmer test scores (mm) at three months. In the Schirmer test, a filter paper strip is placed in the lower eyelid for five minutes and the length of wetting is recorded. Measurements performed with or without anesthesia were eligible.
- Tear film stability, measured as the mean change in tear film break-up time (TBUT; seconds) at three months.
- Frequency of artificial tear use, as defined by trialists, at three months.
- Conjunctival goblet cell density, measured as the mean change from baseline at six months.
- Discontinuation due to adverse events, measured as the proportion of participants who discontinued treatment during the study period.

Search methods for identification of studies

Electronic searches

On 21 January 2025, the Cochrane Eyes and Vision (CEV) Information Specialist searched the following electronic databases for RCTs and controlled clinical trials. We used the MEDLINE trial filter developed by Glanville and colleagues [31]. There were no restrictions on language or year of publication. We conducted a top-up search on 17 March 2026. The search strategies are available in [Supplementary material 1](#).

- Cochrane Central Register of Controlled Trials (CENTRAL; 2025, Issue 1) (which contains the Cochrane Eyes and Vision Trials Register) in the Cochrane Library (searched 17 March 2026)
- MEDLINE Ovid (1946 to 17 March 2026)
- Embase.com (1947 to 17 March 2026)
- PubMed (1948 to 17 March 2026)
- Latin American and Caribbean Health Sciences Literature Database (LILACS) (1982 to 17 March 2026)
- US National Institutes of Health Ongoing Trials Register ClinicalTrials.gov (searched 17 March 2026)
- World Health Organization (WHO) International Clinical Trials Registry Platform (ICTRP) (www.who.int/ictpr/en/; searched 17 March 2026)

Searching other resources

We searched the reference lists of included studies for additional trials. We did not conduct manual searches of conference proceedings or abstracts specifically for this review. Prior to publication, we searched MEDLINE and the Retraction Watch database to ensure that none of the included studies were withdrawn owing to errors or fraud.

Data collection and analysis

When a review author had direct involvement in the conduct, analysis, or publication of a study that could be included in the review, that author was not involved in the eligibility assessment, data extraction, risk of bias assessment, and GRADE assessment of that study.

Selection of studies

Two review authors (SP, CCT, CN, CH, or Cochrane Eyes and Vision methodologist) independently assessed the titles and abstracts of records identified by the search. Each author classified each record as (1) relevant, (2) possibly relevant or unclear, or (3) definitely not relevant (see Criteria for considering studies for this review). For all records classified as (1) or (2) after adjudication, we obtained the full-text reports, and two review authors (SP, CCT, CN, CH, or Cochrane Eyes and Vision methodologist) independently assessed each full-text report for inclusion in the review. We documented the reasons for excluding studies after reviewing the full-text reports. We resolved any discrepancies through discussion.

Data extraction and management

Two review authors (SP, SS, CCT, CN, CH, or Cochrane Eyes and Vision methodologist) independently extracted data for each included study in Covidence [32]. We extracted data relevant to study methods, participants, interventions, and outcomes. We resolved any discrepancies through discussion to reach consensus. One review author entered data into Review Manager [33], and a second review author verified the data entry.

Risk of bias assessment in included studies

We continued using the original risk of bias (RoB) assessment tool to maintain methodological consistency and alignment with the 2019 review [24, 34]. Two review authors (SP, SS, CCT, CN, CH, or Cochrane Eyes and Vision methodologist) independently assessed the risk of bias for each included study according to methods described in the 2011 version of the *Cochrane Handbook for Systematic Reviews of Interventions* [35]. We assessed the following risk of bias domains:

- selection bias (sequence generation and allocation concealment);
- performance bias (masking of participants and study personnel);
- detection bias (masking of outcome assessors);
- attrition bias (loss to follow-up and intention-to-treat analysis); and
- reporting bias (selective outcome reporting)

We judged each domain as low, unclear, or high risk. We resolved any discrepancies through discussion to reach consensus.

Measures of treatment effect

We reported dichotomous outcomes as risk ratios (RRs) with 95% confidence intervals (CIs).

We reported continuous outcomes as mean differences (MDs) with 95% CIs when studies used the same measurement scale. When studies used different instruments to measure the same outcome, we used the standardized mean differences (SMDs) with 95% CIs. We analyzed change scores and final values separately when calculating SMDs, in accordance with guidance in the *Cochrane Handbook for Systematic Reviews of Interventions* [36]. We interpreted SMDs using Cohen's criteria, where 0.2 represents a small effect, 0.5 a moderate effect, and 0.8 a large effect. When change-from-baseline data were unavailable, we used the difference in mean values at specified follow-up time points for the analysis.

Unit of analysis issues

When one eye per individual was randomized, the unit of analysis was straightforward (i.e. individual or eye). When both eyes were randomized, either to the same intervention or different interventions, the analysis had to account for the correlation between two eyes within the same individual (i.e. intra-person correlation). We assessed whether trials (including paired-eye designs and cluster-randomized designs) had properly accounted for the intra-person or intra-cluster correlation, following the recommendations in the *Cochrane Handbook* [37].

Dealing with missing data

We attempted to contact 10 primary study investigators for missing data related to study characteristics and outcome data on two occasions within a two-week period (Agarwal 2019 [38]; Barreto 2009 [39, 40]; Eom 2023 [41, 42]; Jo 2022 [43, 44]; Peng 2021 [45]; Rajpoot 2022 [46, 47]; Rhim 2022 [48, 49]; Shin 2021 [50, 51]; Wirta 2019 [52, 53]; Xu 2023 [54]). We received a response only from the investigator for Rajpoot 2022. For all other studies, we used the available published data. We did not impute missing data for this review.

Reporting bias assessment

We assessed selective outcome reporting as part of the risk of bias assessment. When more than 10 studies were included in a meta-analysis, we assessed potential publication bias using funnel plots.

Synthesis methods

When sufficient data were available (e.g. an effect estimate and its precision) and pooling was considered appropriate, we combined study results in a meta-analysis using Review Manager [33]. We did not pool results when there was evidence of substantial or considerable statistical heterogeneity or when the direction of treatment effects was inconsistent across studies.

We used a random-effects model when three or more studies were included and a fixed-effect model when fewer than three studies were included. For random-effects meta-analyses, we used restricted maximum likelihood (REML) estimation and calculated CIs for the summary effect using the Wald method, in accordance with the updated guidance in Chapter 10 of the *Cochrane Handbook* [36]. For multi-arm studies, we followed guidance in Chapter

23 of the *Cochrane Handbook*, splitting the control group when appropriate [55].

We followed the recommendations in Chapter 12 of the *Cochrane Handbook* when meta-analysis was not possible and qualitatively synthesized the results [56].

Investigation of heterogeneity and subgroup analysis

We assessed clinical and methodological heterogeneity by comparing the design, populations, interventions and comparisons, outcomes, and risk of bias across trials. We assessed statistical heterogeneity using the I^2 statistic, and followed Cochrane guidance on its interpretation [36].

We performed subgroup analyses based on different formulations of CsA (e.g. nanoemulsion, gel-based preparations, etc.).

Equity-related assessment

We did not investigate health inequities in this review.

Sensitivity analysis

We conducted sensitivity analyses to assess the impact of excluding studies at high risk of bias.

Certainty of the evidence assessment

We prepared summary of findings tables for all six comparisons. Two review authors (of SP, SS, and the Cochrane Eyes and Vision methodologist) independently graded the certainty of evidence for the seven outcomes specified in [Outcome](#)

[measures](#) using GRADE criteria (www.gradeworkinggroup.org/). We resolved any discrepancies through discussion to reach consensus within the review team. We downgraded the certainty of the evidence for limitations due to risk of bias, indirectness, inconsistency (unexplained heterogeneity), imprecision (wide confidence intervals), and publication bias.

Consumer involvement

We did not involve consumers in this review update.

RESULTS

Description of studies

Results of the search

For this update, we ran the electronic searches in January 2025, yielding 2206 records. We also identified seven additional records through searching other sources and 740 records through a top-up search on 17 March 2026. After excluding duplicates from the 2953 records retrieved, we had 2795 records for title and abstract screening. We excluded 2611 records, leaving 184 references for full-text review.

After full-text screening, we included 58 RCTs, comprised of 26 studies from the 2019 review (four of which were ongoing when the 2019 review was submitted that have now been published) and 32 newly included studies in this update). We included 40 RCTs in the meta-analyses. We excluded 75 studies, listed 10 studies as awaiting classification, and identified 41 ongoing studies. The PRISMA study flow diagram is shown in [Figure 1](#).

Figure 1. PRISMA study flow diagram

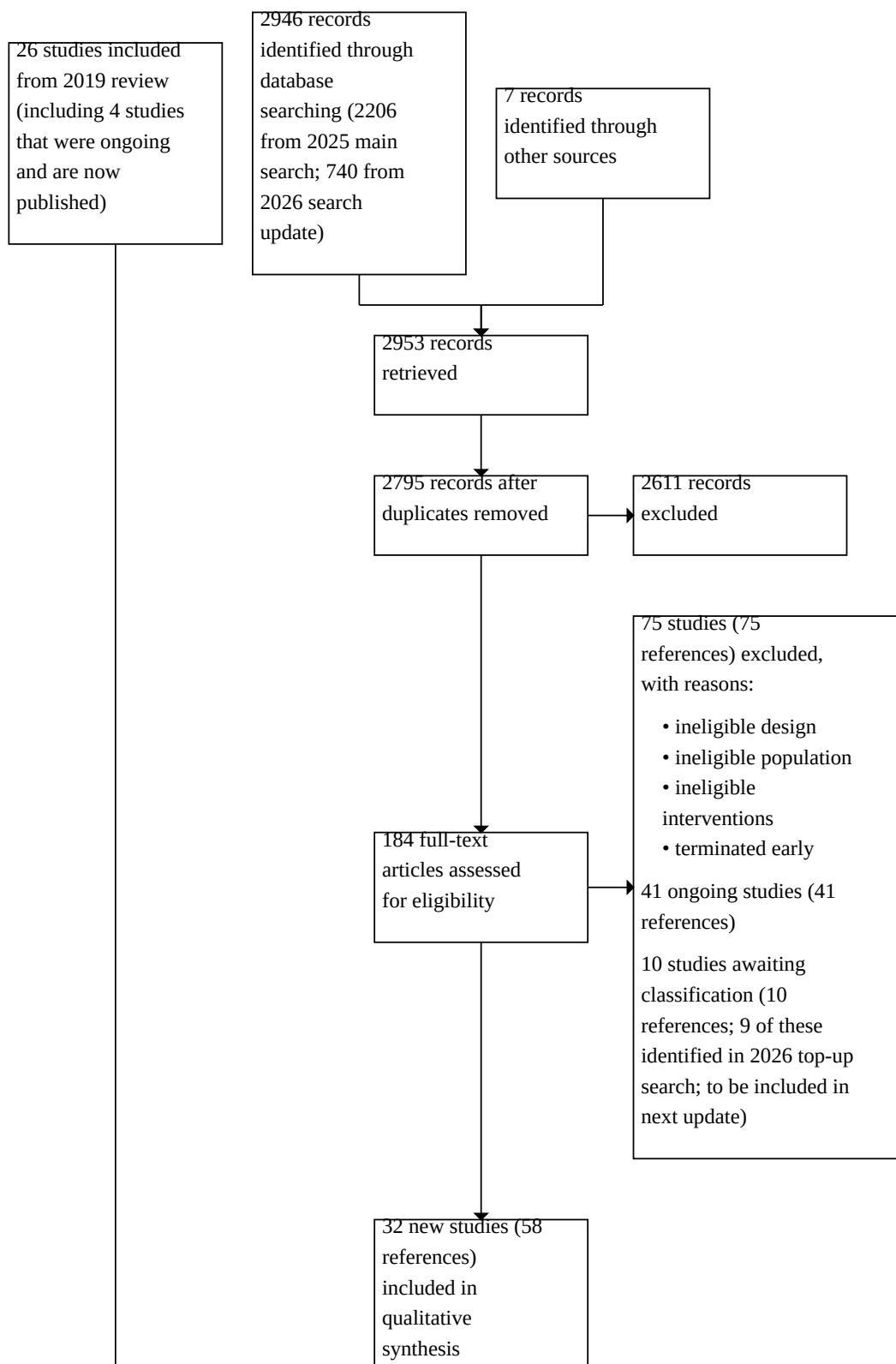
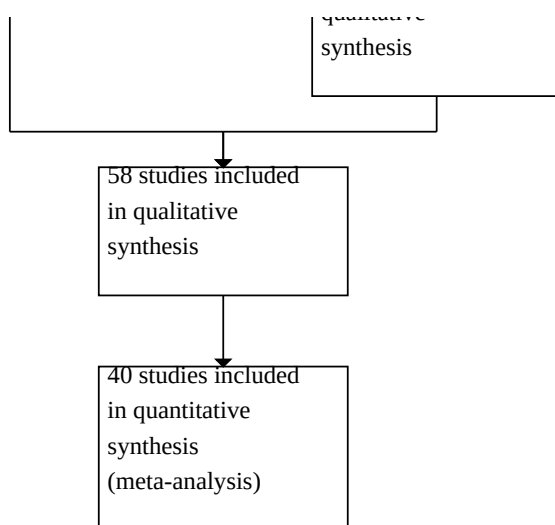


Figure 1. (Continued)



Included studies

In this review update, we included 58 RCTs enrolling 10,225 participants, ranging from 20 (Jain 2007 [57]) to 877 participants (Sall 2000 [58, 59, 60, 61, 62, 63, 64, 65]) per study, with a median of 85 participants (interquartile range [IQR] 51 to 207). Of the 58 included studies, 53 used randomization at the participant level, while four used a paired-eye design with randomization by eye, and one study was unclear. The parallel-group studies typically analyzed outcomes using one eligible eye per participant, often selecting the eye with worse dry eye severity, worse corneal fluorescein staining score, or lower Schirmer test score. None of the studies using the paired-eye design appropriately accounted for the intra-individual correlation between eyes in the analysis. The included studies were conducted around the world: South Korea (13), the USA (12), China (10), India (four) and one each in Brazil, Indonesia, Mexico, Thailand, and Turkey. Three studies were conducted in a multi-country setting in Europe (Baudouin 2017 [66]; Brignole 2001 [67, 68, 69, 70, 71]; Leonardi 2016 [72, 73, 74, 75, 76, 77]). Eleven studies failed to report where the study was conducted. Follow-up periods ranged from three days to 12 months; just over half of studies reported a 3-month (12-week) follow-up (31 studies, 53%). Twenty-eight (48%) studies were funded by industry, while 10 studies received research grants or support from a non-profit organization or foundation; six studies did not receive any funding; and the remaining 14 studies did not report any source of funding. A summary of interventions and comparisons, study design, participant characteristics, and follow-up periods is shown in Table 1. Key study characteristics and study details for all 58 included studies are available in Supplementary material 2.

Participants

The mean age of participants in the included studies ranged from 20 years (Xu 2023) to 68 years (Chung 2013 [78]), with a median of 53 years (IQR 46 to 59). The proportion of women among the participants ranged from 46% (Jo 2022) to 100% (Kang 2020 [79]) across studies (median 80%, IQR 72% to 86%).

Participants were primarily diagnosed with DED based on standardized clinical criteria, including Schirmer test scores (< 10 mm/5 minutes without anesthesia), tear film break-up time (≤ 10 seconds), corneal fluorescein and conjunctival lissamine green or rose Bengal staining, and DED symptoms (e.g. dryness, photophobia, foreign body sensation, fluctuating vision).

Subtypes, etiology, and severity of DED varied widely and included primary and secondary Sjögren's disease, thyroid-associated orbitopathy, meibomian gland dysfunction (MGD), post-laser in situ Keratomileusis (LASIK) DED, post-cataract surgery DED, contact lens-associated DED, human immunodeficiency virus (HIV)-related DED, and DED secondary to anti-glaucoma medications containing benzalkonium chloride-preserved eye drops.

Common exclusion criteria across studies included active ocular infections or non-DED-related inflammation, recent ocular surgery (within three to six months), current or recent use of contact lenses (in studies not assessing the effect of CsA in contact lens-related DED), DED related to cicatricial ocular surface diseases (Steven-Johnson syndrome, ocular cicatricial pemphigoid, chemical burns), use of other anti-inflammatory agents such as lifitegrast, corticosteroids, or other ocular treatments (studies used various time periods before recruiting into the CsA trial; i.e. four weeks to six months), pregnant and breastfeeding women, and systemic medications affecting tear production.

Overall, baseline demographic and clinical characteristics were well-balanced between treatment groups. Of the 58 included studies, 52 reported baseline characteristic equivalence. The remaining six studies did not report or show imbalances (Baiza-Durán 2010 [80]; Kim 2023 [81, 82]; Laibovitz 1993 [83]; NCT01319773 [84]; Peng 2021; Priani 2023 [85]).

Interventions and comparators

Thirty-four studies compared topical CsA 0.05% with vehicle or artificial tears. Of these, 13 studies compared topical CsA 0.05% with vehicles (Baiza-Durán 2010; Brignole 2001; Chen 2010 [86]; Chen 2019 [87]; Guzey 2009 [88, 89]; Liew 2012 [90, 91]; Ma 2015

[92]; NCT04734197 [93]; Peng 2022 [94, 95]; Sall 2000; Stevenson 2000 [96, 97, 98]; Tauber 2018 [99, 100, 101]; Wirta 2019). In all but three (NCT04734197; Tauber 2018; Wirta 2019), participants in both intervention groups were allowed to use artificial tears as adjunctive treatment. The remaining 21 studies compared topical CsA 0.05% plus artificial tears with artificial tears alone. Three studies used nanoemulsions (Kim 2023; Moon 2024 [102]; Rhim 2022), and one study each used a nanomicellar solution (Tauber 2018), water-free eye drops (Wirta 2019), and a gel-based formulation (Peng 2022).

Fourteen studies compared topical CsA 0.05% with artificial tears, placebo, vehicle, or other formulations of CsA plus artificial tears. Nine studies compared CsA 0.05% (Restasis) with CsA 0.1% cationic emulsion (Jung 2023 [103]), CsA 0.05% nanoemulsion (Jo 2022; Kang 2020; Kim 2017 [104]; Park 2019 [105, 106]), CsA 0.08% nanoemulsion (Eom 2023), CsA 0.05% nanoparticulate formulation (Rao 2023 [107, 108]), gel-based formulation (Peng 2021), or trehalose-added formulation (Shin 2021). The remaining five studies compared non-specified CsA 0.05% with CsA 0.1% (Baiza-Durán 2010; Brignole 2001; Rajpoot 2022; Sall 2000), or compared CsA 0.05%, 0.1%, 0.2%, and 0.4% in a 4-arm study (Stevenson 2000). All 14 studies used artificial tear eye drops as adjunctive therapy.

Eleven studies compared topical CsA 0.1% with placebo or vehicle. Two studies compared topical CsA 0.1% cationic emulsion with its vehicle in participants with moderate-to-severe DED (Baudouin 2017) and severe DED (Leonardi 2016). Four studies assessed the benefits and harms of a water-free formulation of CsA 0.1% compared with its vehicle in DED without the use of adjunctive artificial tears (Akpek 2023 [109, 110, 111]; Peng 2024 [112, 113]; Sheppard 2021 [114, 115, 116]; Wirta 2019). Two studies evaluated a preservative-free aqueous nanomicellar formulation of CsA 0.09% with its vehicle (Goldberg 2019 [117, 118, 119]; Tauber 2018). Three additional studies compared CsA 0.1% with its vehicle in Sjögren's disease-associated keratoconjunctivitis sicca (Fan 2003 [120]), DED secondary to cataract surgery (Kudryar 2018 [121]), or moderate-to-severe DED (Stevenson 2000). Adjunctive artificial tear eye drops were applied in all studies.

One study compared CsA 0.1% (Ikervis) with the combination of CsA 0.1% (Ikervis) and sodium hyaluronate in moderate-to-severe DED (Priani 2023). Two studies compared CsA 1% with placebo (Laibovitz 1993; Shen 2020 [122]). Artificial tears were concurrently administered in the intervention groups as adjunctive therapy. Two studies compared topical CsA 2% with placebo or vehicle in participants with secondary Sjögren's disease (Gündüz 1994 [123]), and participants with acquired primary lachrymal disease or Sjögren's disease (Jain 2007). Artificial tears were applied as an adjunctive treatment.

Outcomes

Symptoms of dry eye

Fifty-one of the 58 (88%) included studies assessed symptoms of DED as either a primary or secondary outcome measure. The most commonly used tool was the Ocular Surface Disease Index (OSDI), reported in 33 studies (Barreto 2009; Baudouin 2017; Chen 2019; Chung 2013; Eom 2023; Gao 2023 [124]; Guzey 2009; Hao 2024 [125, 126]; Jain 2007; Jo 2022; Kang 2020; Kim 2009 [127, 128]; Kim 2017; Kim 2023; Kudryar 2018; Lee 2022 [129, 130]; Leonardi 2016; Ma 2015; Park 2017 [131, 132]; Park 2019; Peng 2024; Priani 2023; Rao 2010

[133, 134, 135, 136, 137, 138, 139]; Rao 2023; Rhim 2022; Salib 2006 [140]; Sall 2000; Schrell 2012 [141]; Sheppard 2021; Stevenson 2000; Willen 2008 [142]; Wirta 2019; Xu 2023).

The studies also used four other instruments to assess dry eye symptoms:

- a visual analogue scale (VAS): a simple metric allowing participants to rate symptom severity on a continuum; reported in four studies (Akpek 2023; Fan 2003; Leonardi 2016; Peng 2021);
- the Symptom Assessment in Dry Eye (SANDE) questionnaire: a validated, 2-question instrument measuring frequency and severity; used in four studies (Goldberg 2019; Jung 2023; Moon 2024; Tauber 2018);
- the Standard Patient Evaluation of Eye Dryness (SPEED) questionnaire: a tool assessing symptom frequency and severity over time; reported in one study (Shin 2021);
- the Dry Eye Scoring System (DESS): assesses the frequency and severity of multiple dry eye symptoms; applied in two studies (Rajpoot 2022; Shen 2020).

Additionally, several studies used unvalidated symptom scores such as numeric rating scales or Likert-type scales. Some employed composite symptom-based endpoints, while a few reported subjective improvement descriptively without specifying the assessment method (e.g. reported "foreign body sensation").

In studies using OSDI, outcomes were typically reported as mean change from baseline at various time points, such as at four, eight, or 12 weeks. Among studies using unvalidated symptom scores, the structure and scale of assessment varied widely, limiting comparability. The heterogeneity in assessment tools and time points contributed to the overall variability in symptom reporting across studies.

Ocular surface dye staining

Forty-nine of the 58 included studies (84%) assessed ocular surface staining, using at least one of corneal fluorescein, conjunctival lissamine green, or conjunctival rose Bengal staining as an outcome measure. All studies evaluated ocular surface dye staining as a secondary outcome measure (Agarwal 2019; Akpek 2023; Baiza-Durán 2010; Barreto 2009; Baudouin 2017; Chen 2010; Chen 2019; Demiryay 2011 [143]; Eom 2023; Fan 2003; Gao 2023; Goldberg 2019; Gündüz 1994; Guzey 2009; Hao 2024; Jain 2007; Jo 2022; Jung 2023; Kang 2020; Kim 2017; Kim 2023; Laibovitz 1993; Lee 2022; Leonardi 2016; Liew 2012; Ma 2015; Moon 2024; Park 2017; Park 2019; Peng 2021; Peng 2022; Peng 2024; Prabhasawat 2012 [144, 145]; Priani 2023; Rajpoot 2022; Rao 2010; Rao 2023; Rhim 2022; Sall 2000; Sall 2006 [146, 147, 148]; Schrell 2012; Shen 2020; Sheppard 2021; Shin 2021; Stevenson 2000; Tauber 2018; Willen 2008; Wirta 2019; Xu 2023), while two studies also included corneal staining as a primary outcome measure (Baudouin 2017; Leonardi 2016).

Among the 49 studies that assessed ocular surface dye staining, 31 studies used at least one validated grading scale. Notably, six studies used two different grading scales for corneal and conjunctival staining (Baudouin 2017; Eom 2023; Leonardi 2016; Peng 2024; Sheppard 2021; Wirta 2019). For example, Eom 2023 and Sheppard 2021 used the National Eye Institute (NEI) scale for corneal staining and the Oxford schema for conjunctival staining. The most commonly used scales were the Oxford ocular surface staining scale (used in 17 instances) and the NEI scale (15 instances).

Four studies used the Van Bijsterveld scale, and one study used the SICCA ocular staining score (OSS).

Eighteen studies that reported using ocular surface dye staining as part of the inclusion criteria or as an outcome measure did not specify a grading method or used an unvalidated method (Agarwal 2019; Baiza-Durán 2010; Barreto 2009; Chen 2010; Demiryay 2011; Fan 2003; Gündüz 1994; Guzey 2009; Hao 2024; Jain 2007; Ma 2015; Prabhasawat 2012; Rao 2010; Salib 2006; Schrell 2012; Shen 2020; Stevenson 2000; Willen 2008).

Twenty-six of 49 studies specified a positive corneal fluorescein and/or conjunctival lissamine green or rose Bengal staining score, along with other DED signs, as part of their inclusion criteria (Akpek 2023; Baudouin 2017; Chen 2019; Gao 2023; Jung 2023; Kang 2020; Kim 2017; Laibovitz 1993; Leonardi 2016; Liew 2012; Moon 2024; Park 2017; Park 2019; Peng 2021; Peng 2022; Peng 2024; Rajpoot 2022; Rao 2023; Rhim 2022; Sall 2000; Sall 2006; Sheppard 2021; Shin 2021; Tauber 2018; Wirta 2019; Xu 2023). Among these, 22 studies employed a threshold corresponding to a moderate to severe staining grade, as defined by the particular grading system used.

In studies that used a validated staining grading scale, changes in staining were commonly reported as the mean change from baseline and as differences between groups. These changes were typically assessed at various follow-up visits, including at four, eight, 12, and 26 weeks.

Aqueous tear production

Fifty of the 58 (86%) included studies assessed aqueous tear production using the Schirmer test, as a secondary outcome (Agarwal 2019; Akpek 2023; Baiza-Durán 2010; Barreto 2009; Baudouin 2017; Chen 2010; Chen 2019; Chung 2013; Demiryay 2011; Eom 2023; Fan 2003; Gao 2023; Goldberg 2019; Gündüz 1994; Guzey 2009; Hao 2024; Jain 2007; Jo 2022; Jung 2023; Kang 2020; Kim 2009; Kim 2017; Kim 2023; Laibovitz 1993; Leonardi 2016; Liew 2012; Ma 2015; Moon 2024; Park 2017; Park 2019; Peng 2021; Peng 2022; Peng 2024; Prabhasawat 2012; Prian 2023; Rajpoot 2022; Rao 2010; Rao 2023; Rhim 2022; Salib 2006; Sall 2000; Sall 2006; Shen 2020; Shin 2021; Stevenson 2000; Tauber 2018; Willen 2008; Wirta 2019; Wu 2009 [149]; Xu 2023). Reported outcomes typically included mean Schirmer test values before and after treatment, with some studies noting statistically significant improvements following cyclosporine use. However, variability in testing conditions and follow-up durations limited direct comparisons across studies.

Tear film stability

Forty-three studies (74%) assessed tear film stability using mean change in tear film break-up time (TBUT) as a secondary outcome (Agarwal 2019; Altiparmak 2010 [150]; Baiza-Durán 2010; Baudouin 2017; Chen 2010; Chen 2019; Chung 2013; Demiryay 2011; Eom 2023; Gao 2023; Gündüz 1994; Guzey 2009; Hao 2024; Jain 2007; Jo 2022; Jung 2023; Kang 2020; Kim 2009; Kim 2017; Kim 2023; Laibovitz 1993; Lee 2022; Leonardi 2016; Liew 2012; Ma 2015; Moon 2024; Park 2017; Park 2019; Peng 2021; Peng 2022; Prabhasawat 2012; Prian 2023; Rajpoot 2022; Rao 2010; Rhim 2022; Schrell 2012; Shen 2020; Shin 2021; Tauber 2018; Willen 2008; Wirta 2019; Wu 2009; Xu 2023).

Frequency of artificial tear use

The use of concomitant artificial tear eye drops during the study period was reported in 29 studies (excluding artificial tears used as part of the assigned intervention) (Agarwal 2019; Baiza-Durán 2010; Baudouin 2017; Brignole 2001; Chen 2010; Chen 2019; Eom 2023; Fan 2003; Gündüz 1994; Guzey 2009; Jung 2023; Kang 2020; Kim 2009; Kim 2017; Kudyar 2018; Laibovitz 1993; Lee 2022; Liew 2012; Park 2019; Peng 2022; Prabhasawat 2012; Rajpoot 2022; Rao 2010; Rao 2023; Salib 2006; Sall 2000; Shen 2020; Shin 2021; Stevenson 2000). The most commonly used compositions were unpreserved carboxymethylcellulose sodium and sodium hyaluronate eye drops. The adjunctive use of these eye drops typically ranged from four to six times per day in most studies or was indicated as needed by the study participant.

Eighteen studies included the use of artificial tear eye drops in their intervention arms, as combination therapy with CsA, in the control arms, or both (Altiparmak 2010; Barreto 2009; Demiryay 2011; Gao 2023; Hao 2024; Jo 2022; Kim 2023; Lee 2022; Ma 2015; Park 2017; Peng 2021; Prian 2023; Rhim 2022; Sall 2006; Schrell 2012; Shen 2020; Shin 2021; Wu 2009). The most common combinations with CsA were unpreserved sodium hyaluronate and sodium carboxymethylcellulose eye drops. These artificial tears were typically administered two to four times daily either in the CsA or in the control arm.

Conjunctival goblet cell density

Seven studies included goblet cell density as a secondary outcome measure (Altiparmak 2010; Demiryay 2011; Guzey 2009; Kang 2020; Kim 2009; Rao 2010; Sall 2000). Most of these studies employed conjunctival impression cytology to obtain conjunctival epithelial cells and goblet cells. Changes in conjunctival goblet cell density (cells/mm²) were reported as the mean change from baseline, and comparisons were made between study groups, typically assessed at follow-up visits at four, eight, and 12 weeks.

Discontinuation due to adverse events

Twenty-seven (47%) studies provided a participant flow diagram or described participant flow, showing the number of participants who discontinued the study due to adverse events (Akpek 2023; Altiparmak 2010; Baudouin 2017; Gao 2023; Goldberg 2019; Gündüz 1994; Jo 2022; Kang 2020; Kim 2009; Kim 2017; Kim 2023; Laibovitz 1993; Lee 2022; Leonardi 2016; Moon 2024; Peng 2021; Peng 2022; Rao 2023; Rajpoot 2022; Sall 2000; Schrell 2012; Sheppard 2021; Shin 2021; Stevenson 2000; Tauber 2018; Wirta 2019; Xu 2023).

Ongoing studies

We identified 41 ongoing studies in this update. These studies are comparing various concentrations or formulations (e.g. nanoemulsion) of CsA with one another, vehicle, or artificial tears. The detailed characteristics of ongoing studies are documented in [Supplementary material 5](#).

Excluded studies

There are a total of 88 excluded studies, with reasons for exclusion given in [Supplementary material 3](#). In summary, we excluded 28 studies with non-RCT study designs, 19 studies with ineligible participants (e.g. did not have dry eye at baseline), and 37 studies with ineligible interventions (e.g. steroid and tacrolimus eye drops). We also excluded four studies that were terminated early due to

feasibility issues such as lack of funding, staff availability, or failure to enroll participants.

Risk of bias in included studies

We documented our risk of bias judgments for each included study in [Supplementary material 2](#), and summarize these judgments across studies in [Figure 2](#) and [Figure 3](#).

Figure 2. Risk of bias summary 1: review authors' judgements about each risk of bias item for each included study

	Random sequence generation (selection bias)	Allocation concealment (selection bias)	Blinding of participants and personnel (performance bias): All outcomes	Blinding of outcome assessment (detection bias): All outcomes	Incomplete outcome data (attrition bias): All outcomes	Selective reporting (reporting bias)
Agarwal 2019	?	?	?	?	+	-
Akpek 2023	+	?	+	+	+	+
Altiparmak 2010	?	?	?	+	-	?
Baiza-Durán 2010	?	?	?	?	?	?
Barreto 2009	?	?	-	?	+	+
Baudouin 2017	?	?	?	?	+	?
Brignole 2001	?	?	?	+	?	?
Chen 2010	+	+	?	+	+	?
Chen 2019	+	?	+	?	+	?
Chung 2013	?	?	?	+	+	?
Demiryay 2011	?	?	-	+	+	?
Eom 2023	+	?	?	+	-	-
Fan 2003	?	?	?	?	-	?
Gao 2023	?	?	+	+	+	?
Goldberg 2019	+	+	+	+	+	+
Gündüz 1994	?	?	?	?	+	?
Guzey 2009	+	?	+	+	?	?

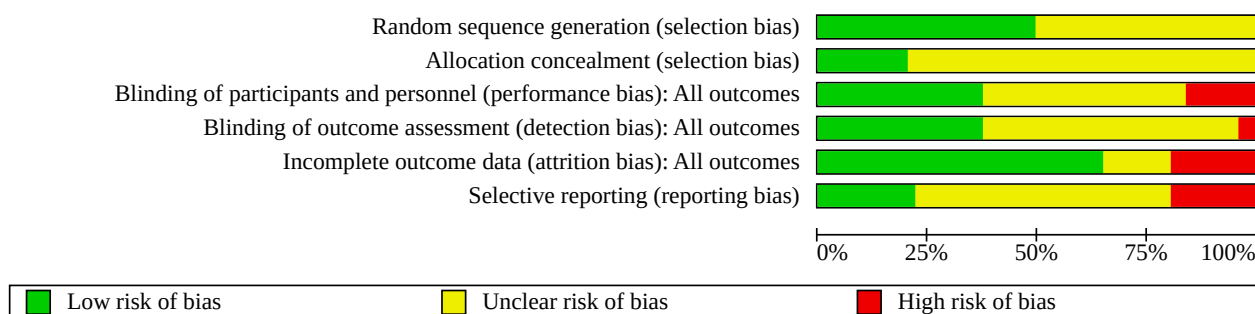
Figure 2. (Continued)

Guzey 2009	+	?	+	+	?	?
Hao 2024	+	?	?	?	+	?
Jain 2007	?	?	?	?	+	?
Jo 2022	?	?	?	?	-	-
Jung 2023	+	+	+	+	-	-
Kang 2020	+	?	?	?	?	?
Kim 2009	+	?	-	?	?	?
Kim 2017	+	?	+	?	+	?
Kim 2023	+	+	?	+	+	+
Kudiyar 2018	+	?	-	?	+	?
Laibovitz 1993	?	?	?	?	-	-
Lee 2022	+	?	-	-	+	-
Leonardi 2016	?	?	?	?	+	+
Liew 2012	?	?	?	?	+	?
Ma 2015	?	?	?	?	?	?
Moon 2024	+	?	+	?	?	?
NCT01319773	?	?	+	+	+	?
NCT04734197	?	?	+	+	+	+
Park 2017	+	?	-	?	+	+
Park 2019	+	+	-	+	-	-
Peng 2021	?	?	?	?	+	?
Peng 2022	+	?	+	+	+	+
Peng 2024	+	+	+	+	+	-
Prabhasawat 2012	+	?	+	?	+	+
Priani 2023	?	?	?	?	+	?
Rajpoot 2022	+	+	+	+	+	?
Rao 2010	+	?	?	?	-	?
Rao 2023	+	+	-	-	+	-
Rhim 2022	+	+	-	+	+	+
Salib 2006	+	?	+	?	+	?
Sall 2000	?	?	+	?	?	?
Sall 2006	+	?	+	?	-	?
Schrell 2012	?	?	?	?	-	?
Shen 2020	+	?	?	?	+	?
Sheppard 2021	+	+	+	+	+	+
Shin 2021	?	?	+	?	+	+

Figure 2. (Continued)

Shin 2021	?	?	+	?	+	+
Stevenson 2000	?	+	+	?	-	-
Tauber 2018	?	?	+	+	+	+
Willen 2008	?	?	+	+	+	?
Wirta 2019	+	+	?	?	+	-
Wu 2009	?	?	?	?	?	?
Xu 2023	?	?	?	?	+	?

Figure 3. Risk of bias summary 2: review authors' judgements about each risk of bias item presented as percentages across all included studies



Random sequence generation

Twenty-nine (50%) studies adequately described suitable methods of random sequence generation, including computer-generated randomization or use of random number tables. We judged these studies to be at low risk of bias for this domain. The remaining 29 (50%) studies did not provide sufficient information on how the random sequence was generated and were thus judged as at unclear risk of bias for this domain.

Allocation concealment

We judged most studies (46 studies, 79%) as having unclear risk of bias for this domain, as they failed to describe how allocation was concealed. Twelve (21%) studies adequately concealed allocation before randomization by using sealed opaque envelopes, an interactive web-based response system, or by central randomization. We judged them at low risk of bias.

Blinding of participants and study personnel

We judged 22 (38%) studies to be at low risk of bias for this domain because they clearly described adequate blinding of participants and study personnel. We assessed nine (16%) studies to be at high risk of bias because they were open-label studies or explicitly stated there was no blinding of participants, personnel, or both. The remaining 27 (47%) studies reported some form of blinding but lacked sufficient detail about who was blinded or how blinding was implemented. We assessed them as having an unclear risk of bias.

Blinding of outcome assessors

Twenty-two (38%) studies adequately blinded outcome assessors; thus, we judged them to be at low risk of bias in this domain. We judged two (3%) studies to be at high risk of bias due to their open-label study design. The remaining 34 (59%) studies either did not mention blinding of outcome assessors or did not specify who was blinded. We judged these studies to have an unclear risk of bias.

Incomplete outcome data

In 38 (66%) studies, outcome data were available for all or nearly all participants, and we judged these studies to be at low risk of bias in this domain. We judged 11 (19%) studies to be at high risk of bias because they had an imbalance in dropouts across intervention groups or there was a high rate of attrition. The remaining nine (16%) studies had incomplete or unclear reporting of attrition, lacked flow diagrams, or omitted reasons for dropout; we judged them to have an unclear risk of bias.

Selective outcome reporting

Thirteen (22%) studies had published protocols or trial registrations, and all prespecified outcomes were reported in the final report; we judged them to have a low risk of bias in this domain. In 11 (19%) studies, outcomes were inconsistently reported between the trial registry and the publication. We assessed these studies as having a high risk of bias. We judged the remaining 34 (59%) studies to have an unclear risk of bias, as the protocol or trial registry was not publicly available.

Synthesis of results

We grouped the 58 included studies into the following six comparisons, based on the comparisons and concentrations/formulations of CsA used:

- topical CsA 0.05% versus vehicle or artificial tears (34 studies);
- other formulations or concentrations of topical CsA versus CsA 0.05% (14 studies);
- topical CsA 0.1% versus vehicle (11 studies);
- topical CsA 0.1% versus topical CsA 0.1% plus artificial tears (1 study);
- topical CsA 1% versus vehicle (2 studies);
- topical CsA 2% versus vehicle (2 studies).

Six multi-arm studies contributed to more than one comparison (Baiza-Durán 2010; Brignole 2001; Sall 2000; Stevenson 2000; Tauber 2018; Wirta 2019). Artificial tears were adjunctively applied in both arms in most studies (81%).

Comparison 1. Topical CsA 0.05% versus vehicle or artificial tears

Thirty-four studies evaluated this comparison. Thirteen studies compared topical CsA 0.05% with vehicle (Baiza-Durán 2010; Brignole 2001; Chen 2010; Chen 2019; Guzey 2009; Liew 2012; Ma 2015; NCT04734197; Peng 2022; Sall 2000; Stevenson 2000; Tauber 2018; Wirta 2019). Participants in both intervention groups in all but three of these studies were allowed to use artificial tears as adjunctive treatment (NCT04734197; Tauber 2018; Wirta 2019). The remaining 21 studies compared topical CsA 0.05% plus artificial tears with artificial tears alone. Warm compress was used for both arms in Rhim 2022. Different formulations of CsA were introduced in the newly included studies in this update: three studies used nanoemulsions (Kim 2023; Moon 2024; Rhim 2022), and one study each used a nanomicellar solution (Tauber 2018), a water-free formulation (Wirta 2019), and a gel-based formulation (Peng 2022).

The underlying etiologies of DED varied considerably across studies and included thyroid orbitopathy (Altıparmak 2010), primary or secondary Sjögren's disease (Gao 2023), severe trachoma (Guzey 2009), anti-glaucoma or preserved medication use (Kim 2023), HIV infection (Barreto 2009), cataract surgery (Chung 2013), LASIK (Salib 2006), meibomian gland dysfunction (Prabhasawat 2012; Rhim 2022), and contact lens wear (Willen 2008). In the paired-eye design, Chung 2013 randomly assigned CsA 0.05% in one eye and normal saline in the other eye. However, this intra-individual comparative study did not account for the non-independence of eyes in its analysis.

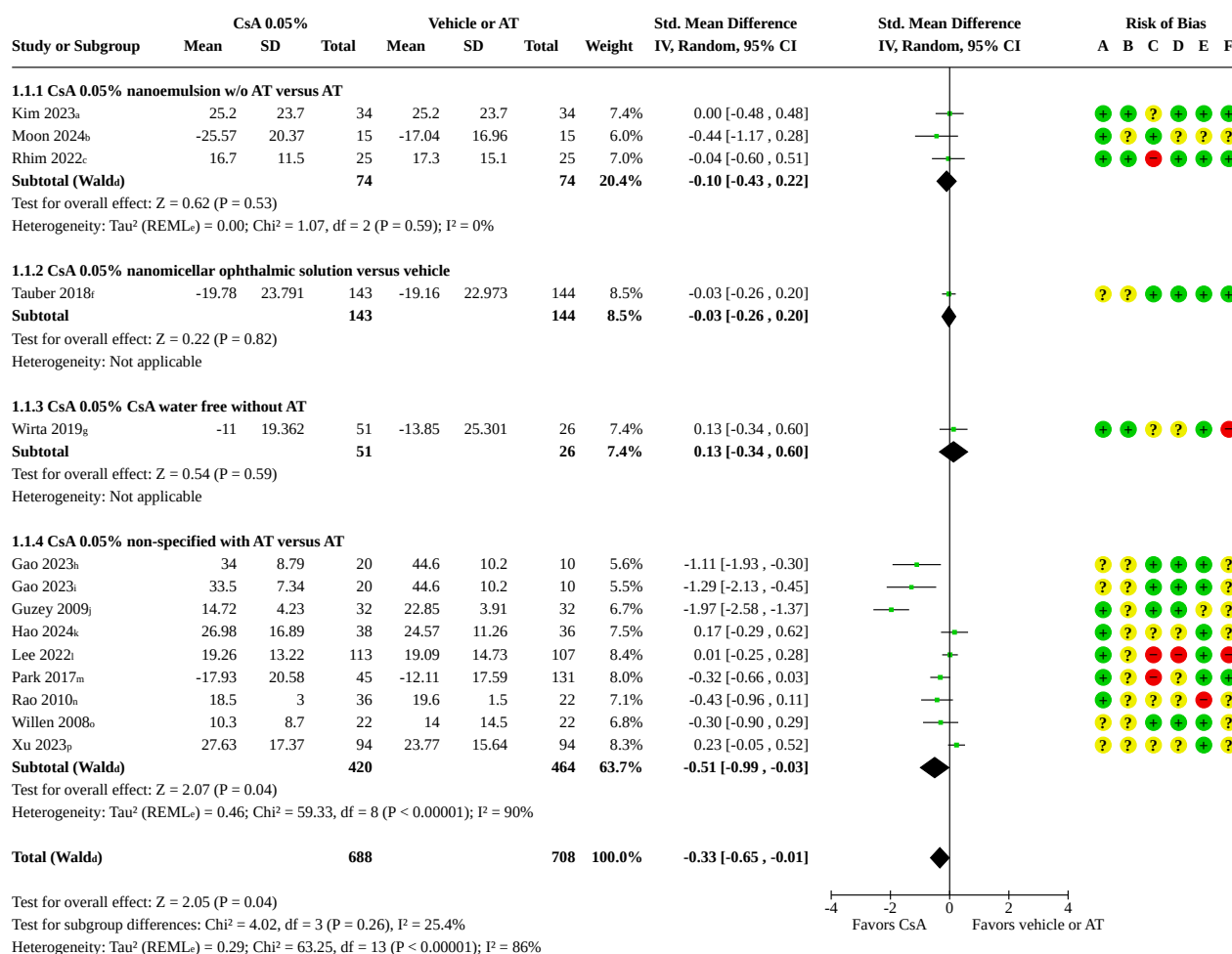
We summarized the critical and important outcomes in [Summary of findings 1](#).

Critical outcome

Subjective symptoms of dry eye at three months

Overall, 27 studies examined the subjective symptoms of DED, and 13 studies reported comparative data and were included in the meta-analysis. All of them used either the OSDI or SANDE symptom questionnaire, both ranging from 0 (no symptoms) to 100 (severe symptoms).

The pooled estimate of these 13 studies suggested that topical CsA 0.05% may improve symptoms of DED at three months compared with vehicle or artificial tears alone (SMD -0.33, 95% CI -0.65 to -0.01; 13 studies, 1396 participants; [Figure 4](#)). However, the results need to be interpreted with caution as considerable heterogeneity was observed. Subgroup analysis based on the CsA eye drop formulation showed that the evidence was imprecise for nanoemulsion (SMD -0.10, 95% CI -0.43 to 0.22; $I^2 = 0\%$; 3 studies, 148 participants; [Figure 4](#)), nanomicellar solution (SMD -0.03, 95% CI -0.26 to 0.20; I^2 not applicable; 1 study, 287 participants; [Figure 4](#)), water-free formulation (SMD 0.13, 95% CI -0.34 to 0.60; I^2 not applicable; 1 study, 77 participants; [Figure 4](#)), and conventional CsA 0.05% ophthalmic solution (the pooled estimate is not shown due to considerable heterogeneity, with an I^2 of 88%).

Figure 4. Forest plot of comparison: CsA 0.05% versus vehicle or artificial tears. Outcome: symptoms of dry eye**Footnotes**

- ^aCsA 0.05% nanoemulsion versus 0.15% hyaluronic acid; both group applied BAK-containing eye drops for glaucoma; paired-eye design; mean at 4 months; OSDI (0-normal to 100-severe); unit of
- ^bCsA 0.05% nanoemulsion (Cyporin N) versus 0.15% HA; mean change from baseline to 3 months; SANDE (0-normal to 100-severe)
- ^cCsA 0.05% nanoemulsion (Cyporin N) + 0.15% HA + warm compress vs 0.15% HA + warm compress; mean at 3 months; OSDI (0-normal to 100-severe)
- ^dCI calculated by Wald-type method.
- ^eTau² calculated by Restricted Maximum-Likelihood method.
- ^fCsA 0.05% nanomicellar ophthalmic solution (OTX-101) vs vehicle; mean change from baseline 84 days; SANDE (0 to 100, higher the worse)
- ^gCsA 0.05% water-free versus vehicle; change from baseline to 14 weeks; OSDI (0-normal to 100-severe); vehicle was divided into two entries (original n=52)
- ^hPreservative-free CsA 0.05% + hyaluronic acid 0.1% versus hyaluronic acid 0.1% only; mean at 12 weeks; OSDI (0-normal to 100-severe); number of hyaluronic acid group was divided by two fr
- ⁱPreservative-free CsA 0.05% versus hyaluronic acid 0.1%; mean at 12 weeks; OSDI (0-normal to 100-severe); number of hyaluronic acid group was divided by two from original (n=20)
- ^jCsA 0.05% (Restasis) + AT versus vehicle + AT; mean at 6 months; OSDI (0-normal to 100-severe)
- ^kCsA 0.05% + 0.1% sodium hyaluronate versus 0.1% sodium hyaluronate; mean at 3 months; OSDI score (0-normal to 100-severe)
- ^lCsA 0.05% (Restasis) + 0.15% HA versus 0.15% HA; mean at 3 months; OSDI (0-normal to 100-severe)
- ^mCsA 0.05% (Restasis) versus HA (combination of 0.1%, 0.15%, 0.3% HA groups); mean change from baseline to 3 months; OSDI (0-normal, 100-severe)
- ⁿCsA 0.05% (Restasis) + AT versus AT; mean at 4 months; OSDI (0-normal to 100-severe)
- ^oCsA 0.05% (Restasis) versus AT; contact wearers; mean at 3 months; OSDI (0-normal to 100-severe)
- ^pCsA 0.05% (Restasis) versus 0.15% hyaluronic acid; mean at 3 months; OSDI (0-normal to 100-severe)

Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)

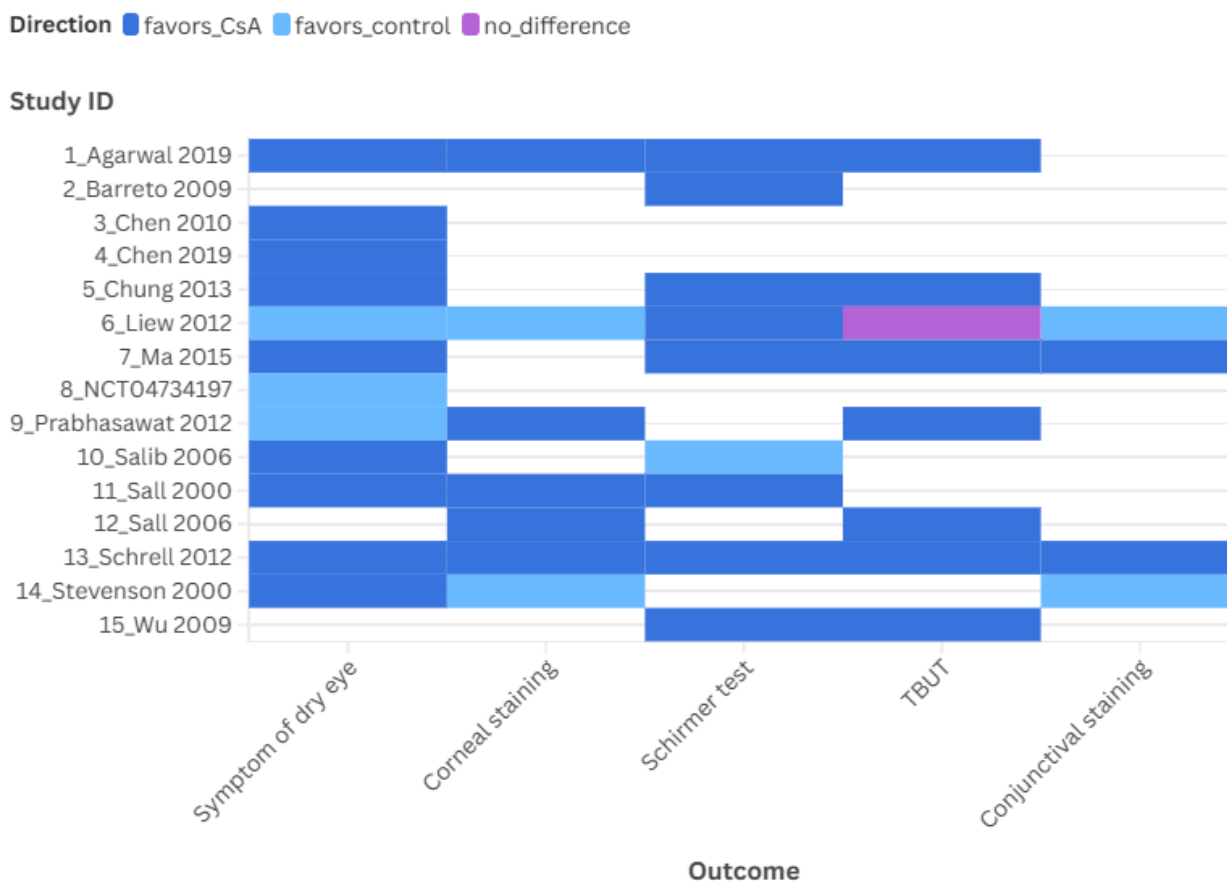
Vote counting based on the direction of effect is described in Figure 5. Nine studies favored the intervention, and three favored artificial tears. The remaining two studies reported no difference

in improvement in OSDI scores between the CsA 0.05% and artificial tears (carboxymethylcellulose 0.5% eye drops) groups at six months in DED secondary to HIV infection. Sall 2006 investigated

two types of artificial tears (i.e. Systane and Refresh), and observed a trend favoring artificial tears (Systane) compared with CsA 0.05% (Restasis) plus artificial tears (Refresh) ($P = 0.0601$). The certainty of

evidence was low, downgraded for risk of bias (−1) and unexplained heterogeneity (−1).

Figure 5. Vote counting based on direction of effect



Important outcomes

Ocular surface dye staining at three months

Twenty-seven studies evaluated ocular surface epithelial disease using dye staining of either the cornea, conjunctiva, or both. Since the grading scale varied across studies, we used the standardized mean difference (SMD). Therefore, we separated the meta-analysis based on mean scores or mean change from baseline scores. For corneal staining, the pooled estimate on mean change scores suggested that CsA 0.05% may improve corneal staining scores slightly compared with vehicle, artificial tears, or both (SMD −0.20, 95% CI −0.31 to −0.09; $I^2 = 0\%$; 7 studies, 1355 participants; Analysis 1.4). Although moderate effects in favor of CsA 0.05% were observed for the estimate on mean scores in corneal staining from 12 studies (Analysis 1.5), we did not combine the study results as considerable heterogeneity was identified ($I^2 = 93\%$). For the remaining studies not included in the meta-analysis, five studies favored CsA 0.05% while two studies favored its vehicle, as shown in Figure 5.

For conjunctival staining, the estimate of effect was imprecise in mean change score (SMD −0.21, 95% CI −0.52 to 0.09; $I^2 = 65\%$; 3 studies, 582 participants; Analysis 1.6). Although two studies suggested that CsA 0.05% may improve conjunctival

staining scores, we did not present the pooled estimates because considerable heterogeneity was identified ($I^2 = 92\%$; Analysis 1.7). Vote counting, based on the direction of effect in four studies not included in the meta-analysis, showed that two studies favored CsA 0.05% and two studies favored vehicle or artificial tears. The remaining two studies reported no difference in evidence between the CsA 0.05% and control groups at three months (Sall 2006) and four months (Barreto 2009). The certainty of evidence was low, downgraded for inconsistency (−1) and risk of bias (−1).

Aqueous tear production at three months

Twenty-eight studies assessed aqueous tear production using Schirmer test scores. We did not present the pooled estimate due to considerable heterogeneity among studies ($I^2 = 98\%$; Analysis 1.8). The formulation subgroup analysis showed that there was no evidence of a difference between a CsA 0.05% nanoemulsion formulation and 0.15% hyaluronic acid (MD 0.64, 95% CI −2.10 to 3.39; $I^2 = 45\%$; 3 studies, 149 participants), while the Schirmer test score favored the CsA 0.05% nanomicellar solution over vehicle (MD 0.49, 95% CI 0.19 to 0.79; 1 study, 287 participants), as well as CsA 0.05% gel-based eye drops (MD 1.40, 95% CI 0.45 to 2.35; 1 study, 627 participants).

The evidence for other unspecified CsA 0.05% eye drops was unclear. We did not pool these data due to the considerable heterogeneity ($I^2 = 98\%$), with the direction of effect varying across studies (seven studies favored CsA, while six studies favored control). For the remaining nine studies, the data were insufficient or incomplete for meta-analysis. Eight studies favored the intervention, while one study favored artificial tears, as shown in [Figure 5](#). The certainty of evidence was low, downgraded for inconsistency (−1) and risk of bias (−1).

Tear film stability at three months

Twenty-nine studies reported tear film stability at three months, measured as the mean change in tear film break-up time (TBUT; seconds). We did not present the pooled estimates of 20 studies (2777 participants) because of considerable methodological heterogeneity and statistical heterogeneity ($I^2 = 95\%$) across studies (Analysis 1.9). Subgroup analysis based on the formulations suggested CsA 0.05% nanoemulsion may improve TBUT compared with vehicle or artificial tears only (MD 2.52, 95% CI 1.00 to 4.03; $I^2 = 78\%$; 3 studies, 149 participants), although the results need to be interpreted with caution. The estimates were imprecise for CsA 0.05% nanomicellar solution (MD 0.04, 95% CI −0.44 to 0.52; 1 study, 287 participants) and gel-based eye drops (MD 0.01, 95% CI −0.34 to 0.37; 1 study, 627 participants). Most of the studies in the conventional CsA 0.05% eye drop subgroup suggested that TBUT favored CsA 0.05% compared with control, but we did not combine the study results due to considerable heterogeneity ($I^2 = 95\%$). For the remaining studies, the direction of effect in seven studies favored CsA 0.05% and one showed no difference ([Figure 5](#)). The certainty of evidence was low, downgraded for inconsistency (−1) and risk of bias (−1).

Frequency of artificial tears use at three months

Three studies reported the frequency of artificial tears use. Chen 2010 reported that there was no between-group difference in adjunctive use of artificial tears during any study period (two to eight weeks). In Liew 2012, the mean artificial tears use was two times daily in both the CsA 0.05% and vehicle groups. Sall 2000 stated that participants in the CsA 0.05% group used artificial tears less frequently than those in the vehicle group ($P = 0.006$). The certainty of evidence was low, downgraded for imprecision (−1) and risk of bias (−1).

Conjunctival goblet cell density at six months

Four studies examined conjunctival goblet cell density over a range of four to 12 months. Although all studies reported that the results favored CsA 0.05%, we did not pool results due to considerable heterogeneity ($I^2 = 97\%$) (Analysis 1.10). The certainty of evidence was low, downgraded for unexplained heterogeneity (−1), and risk of bias (−1).

Discontinuation due to adverse events during the study period

Nine studies reported the number of participants who discontinued the study due to adverse events. The pooled estimate suggested

that treatment with CsA 0.05% probably increases the risk of discontinuing the study due to adverse events (RR 1.99, 95% CI 1.02 to 3.87; $I^2 = 23\%$; 9 studies, 2109 participants, moderate-certainty evidence; Analysis 1.11). Two other studies stated that no participants withdrew due to adverse events in both the CsA 0.05% and control groups (Gao 2023; Xu 2023). The certainty of evidence was moderate, downgraded for risk of bias (−1).

Comparison 2. Other formulations or concentrations of topical CsA versus CsA 0.05%

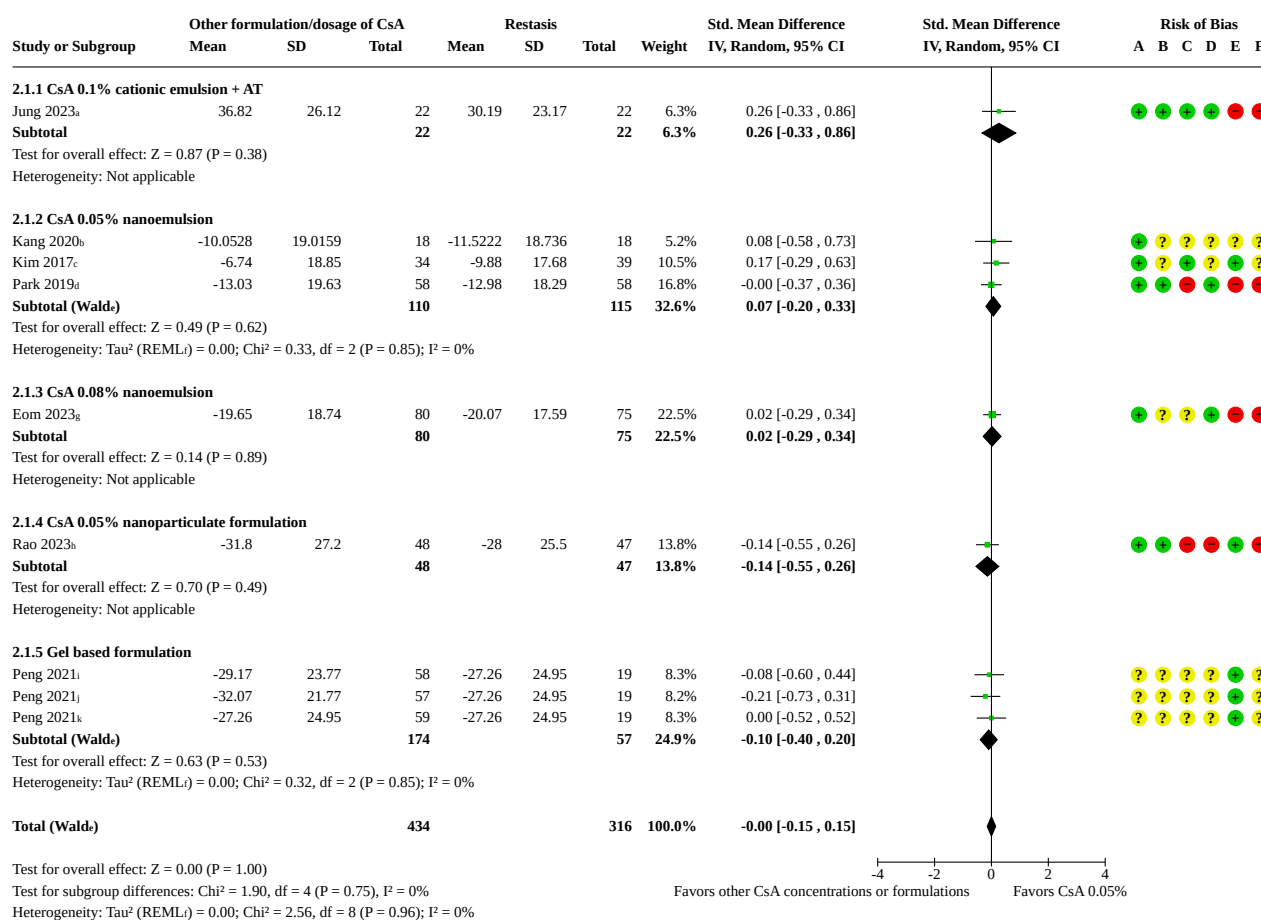
Fourteen studies tested this comparison. Nine newly included studies compared various formulations of CsA to Restasis (CsA 0.05%) as control, as follows: CsA 0.1% cationic emulsion (Jung 2023), CsA 0.05% nanoemulsion (Jo 2022; Kang 2020; Kim 2017; Park 2019), CsA 0.08% nanoemulsion (Eom 2023), CsA 0.05% nanoparticulate formulation (Rao 2023), gel-based formulation (Peng 2021), or trehalose-added formulation (Shin 2021). Four studies compared non-specified CsA 0.05% (i.e. unclear whether it was Restasis or another formulation) with CsA 0.1% (Baiza-Durán 2010; Brignole 2001; Rajpoot 2022; Sall 2000). One study compared CsA 0.05%, 0.1%, 0.2%, and 0.4% in a four-arm design (Stevenson 2000). Artificial tears eye drops were applied as adjunctive therapy in all studies. The studies included participants with Sjögren's disease-associated DED (Kang 2020), MGD (Jo 2022), and moderate-to-severe DED (Brignole 2001; Eom 2023; Rao 2023; Sall 2000; Stevenson 2000). A subset of studies included participants with non-specified types of DED. The follow-up period ranged from three to eight months. We summarized the critical and important outcomes in [Summary of findings 2](#).

Critical outcome

Subjective symptoms of dry eye at three months

Seven studies examined the subjective symptoms of DED using OSDI, SANDE, or a visual analogue scale (VAS) in which scores ranged from 0 (normal) to 100 (severe). The pooled estimate of seven studies comparing various CsA formulations with Restasis was imprecise (SMD −0.00, 95% CI −0.15 to 0.15; $I^2 = 0\%$; 7 studies, 750 participants; [Figure 6](#)), including for: CsA 0.1% cationic emulsion (MD 6.63, 95% CI −7.96 to 21.22; 1 study, 44 participants), CsA 0.05% nanoemulsion (MD 1.27, 95% CI −3.63 to 6.17; $I^2 = 0\%$; 3 studies, 225 participants), CsA 0.08% nanoemulsion (MD 0.42, 95% CI −5.30 to 6.14; 1 study, 155 participants), CsA 0.05% nanoparticulate formulation (MD −3.80, 95% CI −14.40 to 6.80; 1 study, 95 participants), and a gel-based formulation (SMD −0.10, 95% CI −0.40 to 0.20; 1 study, 232 participants). In Baiza-Durán 2010, symptoms such as red eye and dryness favored CsA 0.1% treatment compared with CsA 0.05%, while in another study there was no difference in symptoms between the CsA 0.1% and CsA 0.05% groups (Rajpoot 2022). Three studies did not present between-group differences (Jo 2022; Sall 2000; Shin 2021). We assessed the certainty of evidence as low due to risk of bias (−1) and imprecision (−1).

Figure 6. Forest plot of comparison: other formulations of CsA versus CsA 0.05%. Outcome: symptoms of dry eye



Footnotes

- ^aCsA 0.1% (Ikervis) versus CsA 0.05% (Restasis); mean at 3 months; SANDE score (0-normal to 100-severe)
- ^bCsA 0.05% nanoemulsion (Cyporin N®) versus Restasis; mean change from baseline to 12 weeks; OSDI score (0-normal to 100-severe)
- ^cCsA 0.05% nanoemulsion formulation (ClacierTM) versus Restasis; mean change from baseline to 12 weeks; OSDI score (0-normal to 100-severe)
- ^dCsA 0.05% nanoemulsion (Cyporin N) versus Restasis; mean change from baseline to 12 weeks; OSDI score (0-normal to 100-severe)
- ^eCI calculated by Wald-type method.
- ^fTau² calculated by Restricted Maximum-Likelihood method.
- ^gCsA 0.08% nanoemulsion versus Restasis; mean change from baseline to 32 weeks; OSDI score (0-normal to 100-severe); data extracted from figure
- ^hCsA 0.05% micellar nanoparticle versus Restasis; mean change from baseline at 24 weeks; OSDI (0-normal to 100-severe)
- ⁱCsA gel 0.05% once daily versus Restasis; mean change from baseline at 12 weeks; VAS (0-normal to 100-severe); number of participants in Restasis (n=58) divided by 3
- ^jCsA gel 0.05% twice daily versus Restasis; mean change from baseline at 12 weeks; VAS (0-normal to 100-severe); number of participants in Restasis (n=58) divided by 3
- ^kCsA gel 0.1% versus Restasis; mean change from baseline at 12 weeks; VAS (0-normal to 100-severe); number of participants in Restasis (n=58) divided by 3

Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)

Important outcomes

Ocular surface dye staining at three months

The pooled estimate from five studies showed a trend: newer formulations may slightly improve corneal staining scores, measured as mean change scores (SMD -0.24, 95% CI -0.49 to 0.00; I² = 50%; 5 studies, 572 participants; Analysis 2.2), as well as conjunctival staining scores, measured as mean change scores (SMD -0.17, 95% CI -0.36 to 0.01; I² = 16%; 5 studies, 572 participants; Analysis 2.4). In Jung 2023, there was evidence of no difference in mean corneal staining scores (MD 0.08, 95% CI -0.78 to

0.94; 44 participants; Analysis 2.3) and mean conjunctival staining scores (MD 0.11, 95% CI -1.27 to 1.49; 44 participants; Analysis 2.5) at three months between CsA 0.1% cationic emulsion and CsA 0.05% (Restasis).

For studies not included in the meta-analysis, Jo 2022 reported that corneal staining favored CsA 0.05% nanoemulsion compared with CsA 0.05% among people with MGD at eight weeks, but not at three months. Three studies reported no significant difference between a gel-based CsA formulation (Peng 2021) or other concentrations of CsA (Sall 2000; Stevenson 2000) compared with

CsA 0.05%. However, they reported improvement in lissamine green conjunctival staining scores and corneal fluorescein staining scores with CsA 0.1%. Two studies did not examine between-group differences in ocular staining scores (Baiza-Durán 2010; Shin 2021). We assessed the certainty of evidence as low due to imprecision (−1) and risk of bias (−1).

Aqueous tear production at three months

Although the pooled estimate from five studies suggested that the newer formulations of CsA may improve aqueous tear production (MD 0.30, 95% CI 0.17 to 0.43; $I^2 = 0\%$; 5 studies, 572 participants; Analysis 2.6), individual comparisons such as CsA 0.05% nanoemulsion (MD −0.07, 95% CI −1.23 to 1.09; $I^2 = 0\%$; 3 studies, 225 participants) and CsA 0.05% nanoparticulate formulation (MD 2.30, 95% CI −0.14 to 4.74; 1 study, 192 participants) did not show a significant difference, except for one subgroup, which favored CsA 0.08% nanoemulsion compared with 0.05% CsA (Restasis) (MD 0.30, 95% CI 0.17 to 0.43; 1 study, 155 participants). One study showed that Schirmer test scores favored CsA 0.1% (Rajpoot 2022), whereas four other studies reported no significant difference in aqueous tear production between groups (Baiza-Durán 2010; Peng 2021; Sall 2000; Stevenson 2000). Two studies did not report between-group differences for other formulations of CsA versus CsA 0.05% (Jo 2022; Shin 2021). We assessed the certainty of evidence as low, downgraded for risk of bias (−1) and unexplained heterogeneity (−1).

Tear film stability at three months

The pooled estimate from five studies suggested that newer CsA formulations may not improve tear film stability, assessed as tear film break-up time (TBUT), compared with CsA 0.05% (Restasis) (MD 0.38, 95% CI −0.03 to 0.78; $I^2 = 0\%$; 5 studies, 424 participants; Analysis 2.7). The newer formulations included CsA 0.1% cationic emulsion (MD 0.49, 95% CI −0.99 to 1.97; 1 study, 44 participants) and 0.08% nanoemulsion (MD 0.04, 95% CI −0.60 to 0.68; 1 study, 155 participants). However, the pooled estimate from three of those five studies suggested that CsA 0.05% nanoemulsion may improve TBUT at three months compared with CsA 0.05% (Restasis) (MD 0.62, 95% CI 0.06 to 1.18; $I^2 = 0\%$; 3 studies, 225 participants).

Three other studies not included in the meta-analysis reported that there was no significant difference between groups for TBUT (Jo 2022; Peng 2021; Rajpoot 2022). We assessed the certainty of evidence as low, downgrading for risk of bias (−1) and inconsistency (−1).

Frequency of artificial tears use at three months

Eom 2023 reported the sum of artificial tears use during its 8-week trial period, showing no evidence of a difference between the CsA 0.08% nanoemulsion group and CsA 0.05% (Restasis) group (MD 56.10, 95% CI −5.30 to 117.50; 1 study, 155 participants; Analysis 2.8). The certainty of evidence was low, downgraded for imprecision (−1) and risk of bias (−1).

Conjunctival goblet cell density at six months

The change in goblet cell density grade from baseline to three months showed no difference between CsA 0.05% nanoemulsion and Restasis (MD 0.31, 95% CI −0.51 to 1.13; 1 study, 36 participants; Analysis 2.9). The certainty of evidence was low, downgraded for imprecision (−1) and risk of bias (−1).

Discontinuation due to adverse events during the study period

The pooled analysis of evidence from six studies suggested that participants treated with CsA 0.05% (Restasis) may have a lower risk of discontinuation due to adverse events compared to those receiving other CsA formulations (RR 1.72, 95% CI 1.07 to 2.78; $I^2 = 0\%$; 6 studies, 1421 participants; Analysis 2.10). Four other studies reported that none of the participants withdrew due to adverse events in either group (Jo 2022; Kang 2020; Rao 2023; Shin 2021). The certainty of evidence was moderate, downgraded for risk of bias (−1).

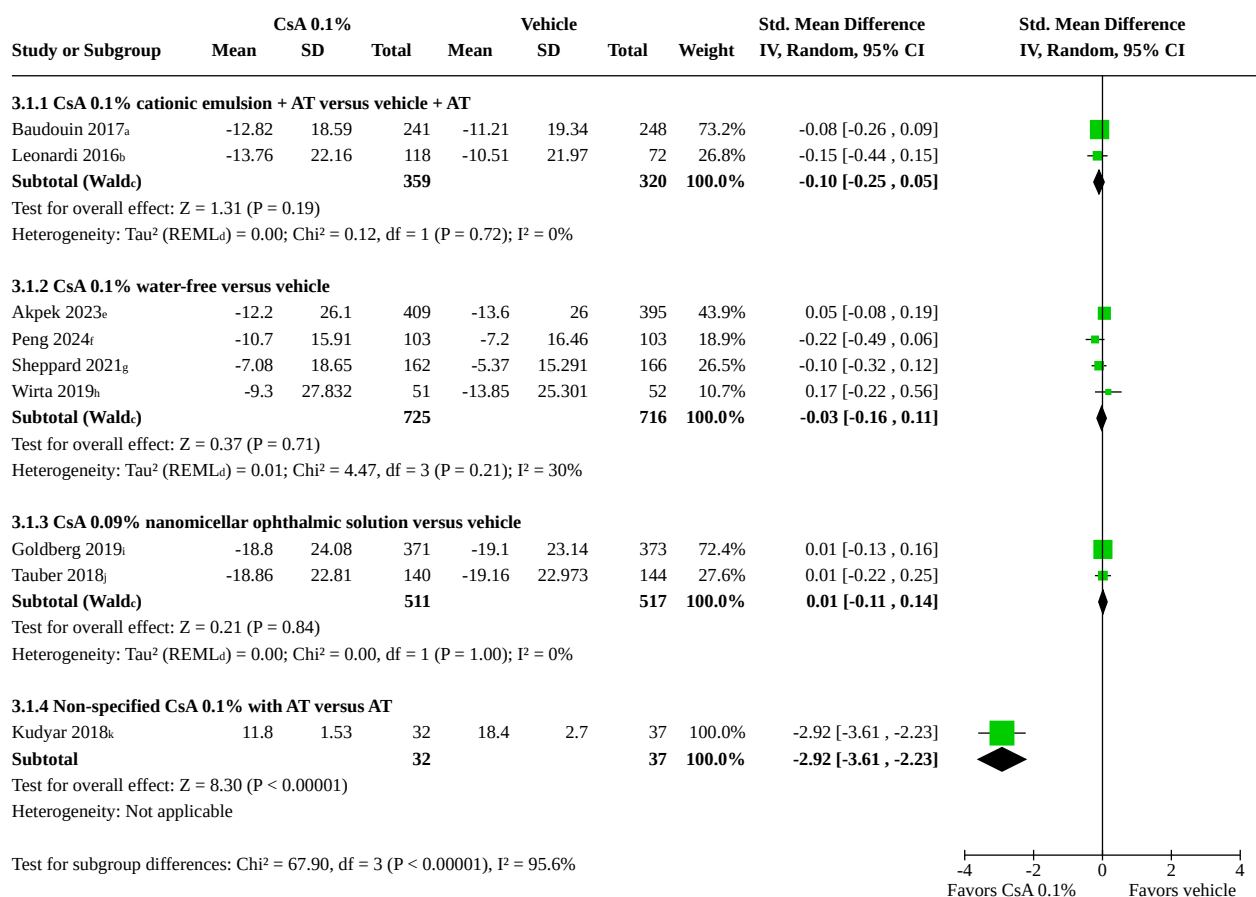
Comparison 3. Topical CsA 0.1% versus vehicle or artificial tears

We included 11 studies in this comparison. Two studies compared topical CsA 0.1% cationic emulsion with its vehicle in participants with moderate-to-severe DED (Baudouin 2017; Leonardi 2016). The follow-up period was six months for both studies, with another six-month open-label treatment for Leonardi 2016. Four studies assessed the benefits and harms of a water-free formulation of CsA 0.1% compared with its vehicle in participants with DED without adjunctive artificial tears. The study durations were: 29 days (Akpek 2023; Peng 2024), three months (Sheppard 2021), and four months (Wirta 2019). Two studies evaluated a preservative-free aqueous nanomicellar formulation of CsA 0.09% with its vehicle during the three-month treatment period (Goldberg 2019; Tauber 2018). Three additional studies compared CsA 0.1% with its vehicle in participants with: Sjögren's disease-associated keratoconjunctivitis sicca (Fan 2003); DED secondary to cataract surgery (Kudryar 2018); and moderate-to-severe DED (Stevenson 2000). Treatment durations ranged from two (Kudryar 2018) to four months (Stevenson 2000). We summarized the critical and important outcomes in [Summary of findings 3](#).

Critical outcome

Subjective symptoms of dry eye at three months

The 11 included studies assessed DED symptoms using the OSDI or SANDE questionnaires, or a visual analogue scale (ranging from 0 to 100, with higher scores indicating worse symptoms). We included these studies in the meta-analysis (Analysis 3.1; [Figure 7](#)). We did not present the pooled estimate from nine studies as substantial heterogeneity ($I^2 = 92\%$) was observed, with direction of effect varying across studies (five studies favored CsA while four studies favored vehicle or artificial tears). Pooled estimates for the formulation subgroup analyses were imprecise for the comparisons of: CsA 0.1% cationic emulsion versus vehicle at six months (MD −1.96, 95% CI −4.94 to 1.02; $I^2 = 0\%$; 2 studies, 679 participants); water-free CsA 0.1% versus vehicle at one month to 14 weeks (MD −0.72, 95% CI −3.36 to 1.93; $I^2 = 26\%$; 4 studies, 1441 participants); and aqueous nanomicellar CsA 0.1% versus vehicle at three months (MD 0.30, 95% CI −2.56 to 3.16; $I^2 = 0\%$; 2 studies, 1028 participants); but not for the comparison of non-specified CsA 0.1% versus vehicle at two months, where the result favored CsA 0.1% over vehicle (MD −6.60, 95% CI −7.62 to −5.58; 1 study, 69 participants). Fan 2003 showed no evidence of a difference at three months between the CsA and vehicle groups. Stevenson 2000 reported that CsA 0.1% showed significant improvement compared with its vehicle at week 14. We downgraded the certainty of evidence from high to low due to risk of bias (−1) and unexplained heterogeneity (−1).

Figure 7. Forest plot of comparison: CsA 0.1% cationic emulsion plus artificial tears versus vehicle plus artificial tears. Outcome: symptoms of dry eye**Footnotes**^aCsA 0.1% cationic emulsion (Ikervis) + AT versus vehicle + AT; OSDI (0-normal to 100-severe); mean change from baseline to 6 months^bCsA 0.1% cationic emulsion + AT (Ikervis) versus vehicle + AT; OSDI (0-normal to 100-severe); mean change from baseline to 6 months^cCI calculated by Wald-type method.^dTau² calculated by Restricted Maximum-Likelihood method.^eCsA 0.1% water-free (CsA Sol, Novaliq GmbH) versus vehicle; least squares mean change from baseline to 29 days; eye dryness score is rated on a visual analogue scale (^fCsA 0.1% water-free (SHR8028) versus vehicle; change from baseline to 29 days; OSDI (0-normal, 100-severe)^gCsA 0.1% water-free versus vehicle; mean change from baseline to 1 month; OSDI (0-normal to 100-severe)^hCsA 0.1% water-free versus vehicle; change from baseline to 14 weeks; OSDI (0-normal to 100-severe)ⁱCsA 0.09% nanomicellar ophthalmic solution versus vehicle; Symptom Assessment in Dry Eye (SANDE) questionnaire (0 to 100, higher the worse); mean change from baseline to 3 months; SANDE (0 to 100, higher the worse)^jCsA 0.09% nanomicellar ophthalmic solution (OTX-101) versus vehicle; mean change from baseline to 3 months; SANDE (0 to 100, higher the worse)^kCsA 0.1% + 0.5% carboxymethylcellulose versus 0.5% carboxymethylcellulose; participants with cataract; mean at 2 months; OSDI (0-normal to 100-severe)**Important outcomes****Ocular surface dye staining at three months**

Nine studies examined corneal fluorescein staining according to the NEI scale or a modified Oxford scale. The pooled estimate of evidence from seven studies showed small effects favoring CsA 0.1% (SMD -0.16, 95% CI -0.24 to -0.08; I² = 15%; 7 studies, 2940 participants; Analysis 3.2). Subgroup analysis by formulation favored CsA 0.1% cationic emulsion (SMD -0.24, 95% CI -0.42 to -0.06; 1 study, 489 participants), water-free CsA (SMD -0.16, 95% CI -0.27 to -0.06; I² = 0%; 4 studies, 1441 participants), and aqueous nanomicellar CsA (SMD -0.14, 95% CI -0.38 to 0.10; I² = 69%; 2 studies, 1010 participants) over vehicle. Although not included in the meta-analysis, Leonardi 2016 showed significant improvement in corneal fluorescein staining by at least two grades

among 80 (51.9%) and 41 (45.1%) participants in the CsA 0.1% cationic emulsion group and vehicle group, respectively. Stevenson 2000 reported that mean scores for superficial punctate keratitis showed an improvement throughout the treatment period of four months, but no significant differences were observed between intervention groups.

Eight studies examined conjunctival staining. The pooled estimate from five studies showed a negligible effect on conjunctival staining in favor of CsA 0.1% (SMD -0.20, 95% CI -0.28 to -0.12; I² = 0%; 5 studies, 2303 participants; Analysis 3.3). Additionally, Sheppard 2021 reported that conjunctival lissamine green staining showed a reduction after one month of treatment with CsA 0.1% over vehicle. In two studies, CsA 0.1% showed a significant reduction in conjunctival staining, but the between-group difference compared

with CsA and vehicle was unclear (Fan 2003; Wirta 2019). We assessed the certainty of evidence as moderate for this outcome, downgrading for risk of bias (−1).

Aqueous tear production at three months

The pooled estimate of three studies indicated that CsA 0.1% may improve aqueous tear production compared with vehicle (MD 0.66, 95% CI 0.09 to 1.24; $I^2 = 30\%$; 3 studies, 735 participants; Analysis 3.4). Schirmer test scores improved over the study period in three studies, but there was no evidence of a difference between groups in two additional studies (Baudouin 2017; Stevenson 2000). The certainty of evidence for this outcome was low due to imprecision (−1) and risk of bias (−1).

Tear film stability at three months

Summary estimates for tear film stability (expressed as TBUT) in a comparison of CsA 0.1% with vehicle showed that the mean change from baseline and at three or six months was imprecise (MD 0.15, 95% CI −0.11 to 0.42; $I^2 = 0\%$; 3 studies, 979 participants; Analysis 3.5). The certainty of evidence for this outcome was low due to imprecision (−1) and risk of bias (−1).

Frequency of artificial tear usage at three months

None of the studies reported this outcome.

Conjunctival goblet cell density at six months

None of the studies reported this outcome.

Discontinuation due to adverse events during the study period

Seven studies reported the number of participants who discontinued the study due to adverse events. Treatment periods ranged from one to six months. Summary estimates indicate that participants treated with CsA 0.1% were more likely to discontinue the study due to adverse events compared with those who were assigned to vehicle groups (RR 1.58, 95% CI 1.04 to 2.41; $I^2 = 0\%$; 7 studies, 3051 participants). The certainty of evidence for this outcome was moderate due to risk of bias (−1).

Comparison 4. Topical CsA 0.1% versus topical CsA 0.1% plus artificial tears

Priani 2023 compared CsA 0.1% (Ikervis) with the combination of CsA 0.1% (Ikervis) and sodium hyaluronate in participants with moderate to severe DED during the 1-month study period. We summarized the critical and important outcomes in [Summary of findings 4](#).

Critical outcome

Subjective symptoms of dry eye at three months

DED symptoms measured using the OSDI questionnaire showed no difference between CsA 0.1% versus CsA 0.1% plus sodium hyaluronate eye drops at one month (MD 1.02, 95% CI −2.26 to 4.30; 1 study, 19 participants; Analysis 4.1). The certainty of evidence was very low due to indirectness (−1) and imprecision (−2).

Important outcomes

Ocular surface dye staining at three months

Priani 2023 examined ocular staining according to the Oxford scale. The authors found that the degree of ocular staining did not indicate a difference between the intervention groups after one

month of treatment. The certainty of evidence was very low due to indirectness (−1) and imprecision (−2).

Aqueous tear production at three months

Aqueous tear production measured by the Schirmer test score was imprecise at one month (MD 0.33, 95% CI −5.39 to 6.05; 1 study, 19 participants; Analysis 4.2). The certainty of evidence was very low due to indirectness (−1) and imprecision (−2).

Tear film stability at three months

Tear film stability (expressed as TBUT) also showed no difference between the CsA and CsA plus artificial tears groups at one month (MD −0.55, 95% CI −2.89 to 1.79; 1 study, 19 participants; Analysis 4.3). The certainty of evidence was very low due to indirectness (−1) and imprecision (−2).

Frequency of artificial tear usage at three months

Priani 2023 did not report this outcome.

Conjunctival goblet cell density at six months

Priani 2023 did not report this outcome.

Discontinuation due to adverse events during the study period

Priani 2023 did not report this outcome.

Comparison 5. Topical CsA 1% versus vehicle or artificial tears

Two studies compared CsA 1% eye drops with vehicle in participants with DED. Artificial tears were concurrently administered in the intervention groups as an adjunctive therapy. The treatment period was eight weeks in Laibovitz 1993 and three months in Shen 2020. Laibovitz 1993, which used a cross-over design, reported that 17 of 26 (65.4%) participants were ineligible for the second phase, and trialists analyzed efficacy data from the first phase only. However, we excluded this study from the meta-analysis because it did not report standard deviations. We summarized the critical and other important outcomes in [Summary of findings 5](#).

Critical outcome

Subjective symptoms of dry eye at three months

Shen 2020 showed that symptoms of DED (as measured with a 0- to 15-point scale, where higher scores indicate worse symptoms) improved in the CsA 1% group after three months of treatment compared with artificial tears (i.e. sodium hyaluronate 0.3%) eye drops (MD −2.34, 95% CI −3.70 to −0.98; 60 participants; Analysis 5.1). Laibovitz 1993 also reported that DED symptoms, especially foreign body sensation, improved in the CsA 1% group compared to the vehicle group. The certainty of evidence was very low due to risk of bias (−1) and imprecision (−2).

Important outcomes

Ocular surface dye staining at three months

Corneal staining (0 = normal to 15 = severe) improved in the CsA 1% group compared with the artificial tears group (MD −0.33, 95% CI −0.56 to −0.10; 1 study, 60 participants) at three months. Laibovitz 1993 reported that rose Bengal staining scores improved nearly 80% in the CsA 1% group and 60% in the vehicle group at the end of the first phase at six weeks. The certainty of evidence was very low due to risk of bias (−1) and imprecision (−2).

Aqueous tear production at three months

In Shen 2020, the mean Schirmer test score at three months favored CsA 1% compared with artificial tears (MD 2.01, 95% CI 1.11 to 2.91; 1 study, 60 participants; Analysis 5.3). In Laibovitz 1993, the mean Schirmer test score was 6 mm/5 minutes in the CsA 1% group and 3 mm/5 minutes in the vehicle group at six weeks. The certainty of evidence was very low due to risk of bias (−1) and imprecision (−2).

Tear film stability at three months

The mean TBUT favored CsA 1% at three months (MD 2.77, 95% CI 2.14 to 3.40; 1 study, 60 participants; Analysis 5.4) in Shen 2020. Laibovitz 1993 also reported a favorable outcome for the CsA 1% group, showing a mean improvement in TBUT of 9.48 seconds from baseline, compared with 2.42 seconds worsening in the vehicle group. The certainty of evidence was very low due to risk of bias (−1) and imprecision (−2).

Frequency of artificial tear usage at three months

Neither study reported this outcome.

Conjunctival goblet cell density at six months

Neither study reported this outcome.

Discontinuation due to adverse events during the study period

Of the 26 participants enrolled in Laibovitz 1993, none of the participants discontinued the study due to adverse events. Shen 2020 did not report adverse events. The certainty of evidence was very low due to risk of bias (−1) and imprecision (−2).

Comparison 6. Topical CsA 2% versus vehicle

Two studies compared topical CsA 2% with its vehicle: participants in Gündüz 1994 had secondary Sjögren's disease-associated DED, while participants in Jain 2007 had acquired primary lachrymal disease (APLD) or Sjögren's disease. Artificial tears eye drops were applied as an adjunctive treatment in both studies. Jain 2007 was a cross-over design without a wash-out period between the two phases, and the carry-over effect was not addressed. Since Jain 2007 did not report separate results by phase, we did not include it in the analysis. We summarized the critical and other important outcomes in [Summary of findings 6](#).

Critical outcome

Subjective symptoms of dry eye at three months

Gündüz 1994 did not assess this outcome. Jain 2007 (30 participants) reported improved symptoms of DED favoring CsA 2%. Trialists reported improvements in symptoms of dryness, gritty sensation, and redness in the CsA 2% group compared to the

placebo group. The certainty of evidence was very low due to risk of bias (−1) and imprecision (−2).

Important outcomes

Ocular surface dye staining at three months

Both studies found no significant differences in the final mean staining scores between treatment groups (Gündüz 1994; Jain 2007). The certainty of evidence was very low due to risk of bias (−1) and imprecision (−2).

Aqueous tear production at three months

Gündüz 1994 found no evidence of a significant difference between CsA 2% and the vehicle group in Schirmer test scores (MD 0.35, 95% CI −0.24 to 0.94; 1 study, 30 participants). The certainty of evidence was very low due to risk of bias (−1) and imprecision (−2).

Tear film stability at three months

Gündüz 1994 reported that CsA 2% might improve TBUT at two months (MD 2.82, 95% CI 2.17 to 3.47; 1 study, 30 participants; Analysis 6.3), but the evidence was very uncertain. In contrast, Jain 2007 found no difference in TBUT between the two groups at two months. The certainty of evidence was very low due to risk of bias (−1) and imprecision (−2).

Frequency of artificial tear usage at three months

Neither study reported this outcome.

Conjunctival goblet cell density at six months

Neither study reported this outcome.

Discontinuation due to adverse events during the study period

In Gündüz 1994, three (20%) eyes in the CsA group and two (13.3%) eyes in the placebo group experienced mild discomfort, but none of the participants discontinued the treatment due to adverse events. The certainty of the evidence was very low due to risk of bias (−1) and imprecision (−2).

Reporting biases

We assessed the risk of selective outcome reporting for each study in the [Risk of bias in included studies](#) section. We created funnel plots for outcomes with 10 or more studies included in the meta-analysis ([Figure 8](#); [Figure 9](#); [Figure 10](#); [Figure 11](#)). Although some asymmetry was observed, especially in the plots for aqueous tear production (Schirmer test) and tear film stability (TBUT), the direction and magnitude of the asymmetry varied across subgroups and outcomes. Due to variation in CsA formulations and the limited number of studies in each subgroup, we cannot conclusively determine the presence of small study effects or publication bias.

Figure 8. Funnel plot assessing publication bias for studies comparing CsA 0.05% versus vehicle or artificial tears only: analysis 1.1 – symptoms of dry eye, mean change from baseline score

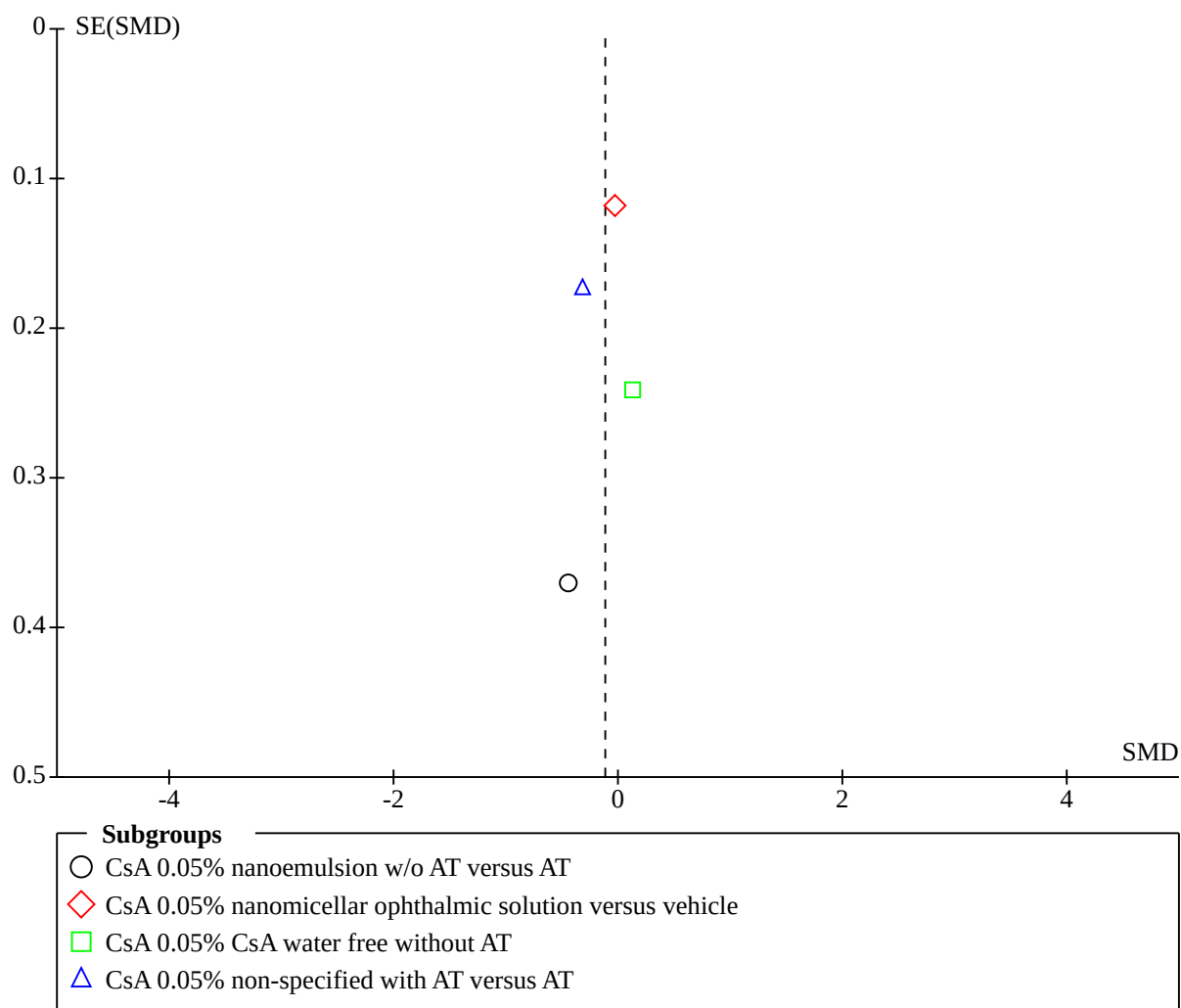


Figure 9. Funnel plot assessing publication bias for studies comparing CsA 0.05% versus vehicle or artificial tears only: analysis 1.5 – corneal staining, mean score

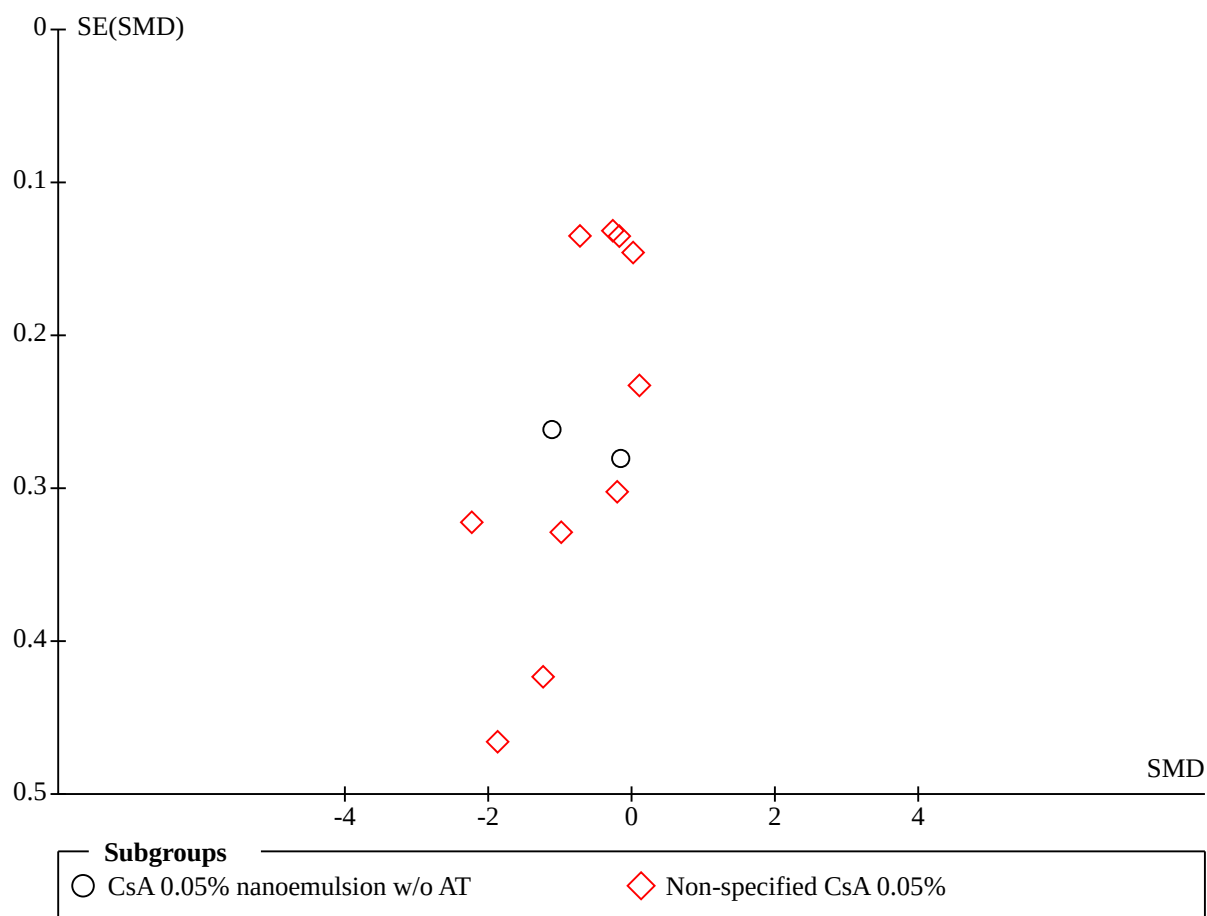


Figure 10. Funnel plot assessing publication bias for studies comparing CsA 0.05% versus vehicle or artificial tears only: analysis 1.8 – Schirmer test score

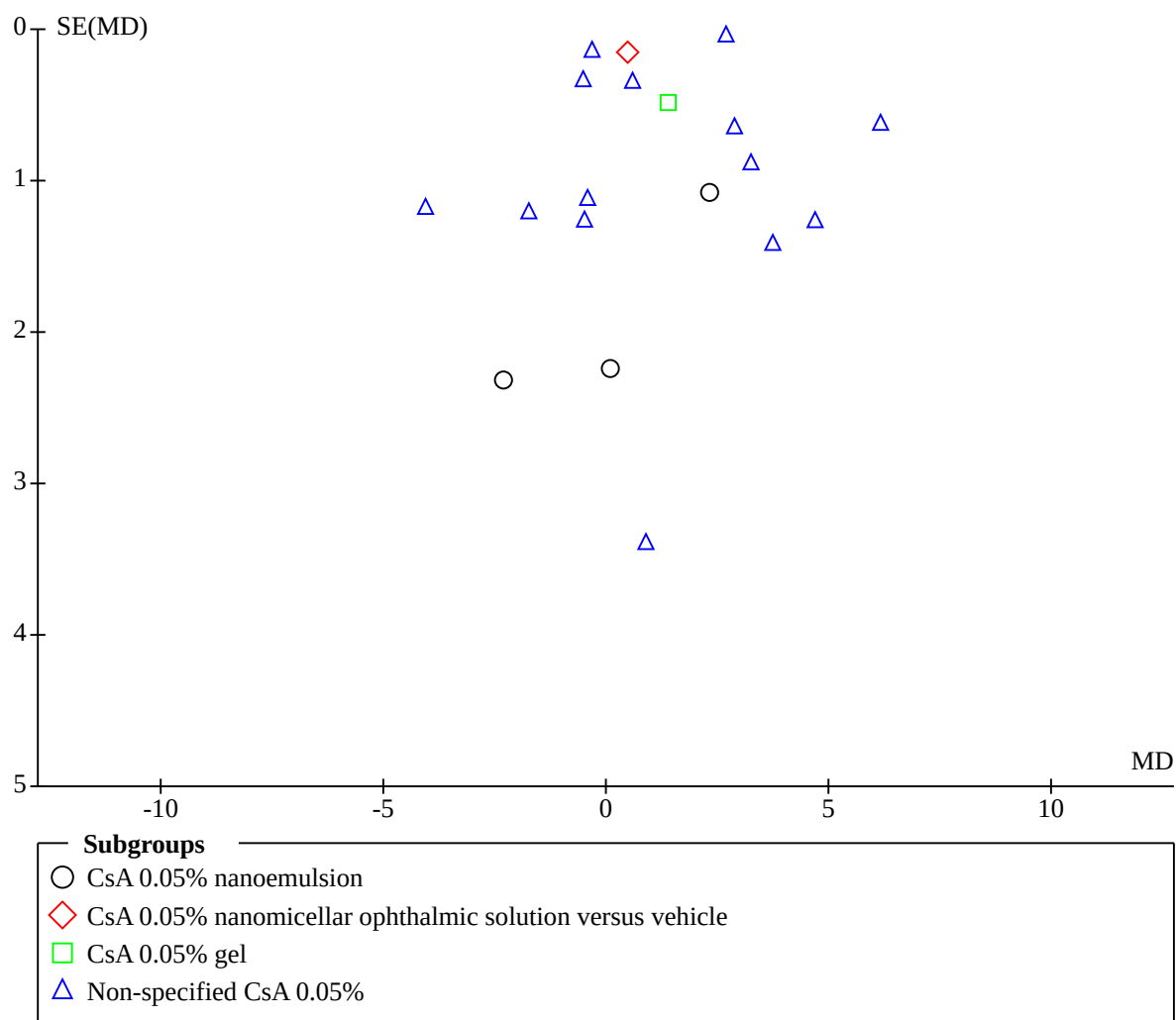
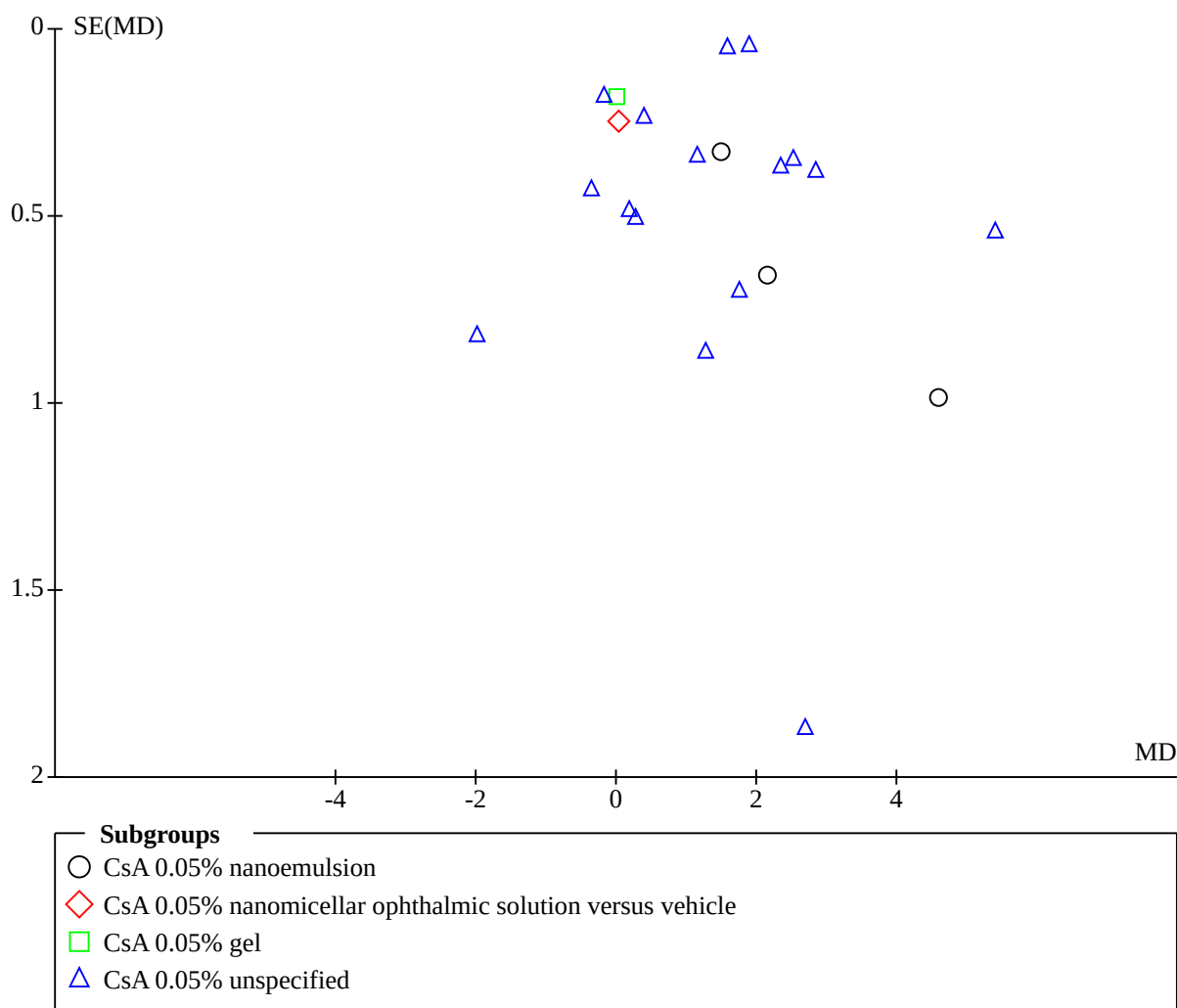


Figure 11. Funnel plot assessing publication bias for studies comparing CsA 0.05% versus vehicle or artificial tears only: analysis 1.9 – tear film break-up time (TBUT)

DISCUSSION

Summary of main results

In this review update, we included 58 RCTs involving 10,225 participants with DED, of which 32 were newly included studies. Most studies compared CsA 0.05% or 0.1% with vehicle or artificial tears, whereas others compared different formulations of CsA (e.g. nanoemulsion and nanomicellar solution) with vehicle, artificial tears, or head-to-head trials with active comparator of CsA 0.05% (e.g. Restasis). Some of the newly included studies investigated newer CsA formulations aimed at improving drug delivery and efficacy in DED. These included 0.05% nanoemulsions, 0.08%, 0.09% nanomicellar formulations, 0.05% gel, 0.1% cationic emulsions, and 0.1% water-free solutions, often compared with the conventional emulsion Restasis (0.05%) or vehicle controls, alongside artificial tears.

We were unable to pool effect estimates across many analyses due to substantial heterogeneity. In most studies, low-certainty evidence suggested that symptoms of DED may improve in

participants treated with CsA 0.05% or CsA 0.1% compared with those who received vehicle or artificial tears. The results were more consistent for ocular surface signs, as measured by corneal and conjunctival staining, where improvements favored CsA over vehicle or artificial tears across studies and comparisons. For aqueous tear production (measured by Schirmer test scores) and tear film stability (measured as tear film break-up time), low-certainty evidence suggested that CsA may improve tear production and tear film stability compared with vehicle or artificial tears across different concentrations and formulations of CsA (i.e. 0.05% and 0.1%). There was limited evidence for the frequency of artificial tear usage and conjunctival goblet cell density as measured by impression cytology across studies.

In short, the pooled analyses in the current review showed statistically significant improvements in several objective and subjective DED parameters favoring CsA over vehicle or artificial tears. However, taken together, the magnitude of benefit appears modest and does not consistently exceed thresholds for clinically meaningful change. These findings suggest that while CsA may

confer measurable biological effects on the ocular surface, the practical impact on clinical signs and symptoms outcomes needs cautious interpretation. The extremely high heterogeneity observed across several pooled analyses (I^2 up to 98%) further limits confidence in the clinical applicability of some results, suggesting that treatment effects may differ across populations, disease severities, and study designs.

Moderate-certainty evidence suggested that CsA 0.05% and CsA 0.1% likely increased the risk of participants discontinuing the study due to adverse events compared with those in the vehicle or artificial tears groups, although attrition rates were generally low across studies (4% to 6% across comparisons). Overall tolerability remains acceptable, and serious safety concerns are not evident.

The chronic course of dry eye disease, together with limited long-term safety and efficacy data, underscores the need for well-designed prospective RCTs of newer cyclosporine formulations with extended follow-up to clarify their sustained therapeutic effects, safety profile, and role in long-term disease management.

Limitations of the evidence included in the review

While we included 58 RCTs, several limitations reduced our confidence in the findings. The overall certainty of the body of evidence ranged from very low to moderate in this review update. For most outcomes in the main comparisons (CsA 0.05% or CsA 0.1% versus vehicle or artificial tears), we downgraded the evidence certainty by two levels to low, primarily due to inconsistency in results across studies, indirectness from variations in outcome assessment time points, and methodological concerns identified through risk of bias assessments.

We were unable to include 18 studies in meta-analyses because of insufficient or incomplete reporting of outcome data, such as missing precision measures. We summarized these studies narratively, which may have limited the precision and interpretability of pooled effect estimates.

This update covers participants with various underlying causes of DED, such as primary and secondary Sjögren's disease, meibomian gland dysfunction (MGD), thyroid-associated orbitopathy, post-LASIK DED, post-cataract surgery DED, contact lens-associated DED, HIV-related DED, and anti-glaucoma medications/preservative-associated DED. This diversity improves the generalizability and clinical applicability of our findings. However, we were unable to perform potentially informative subgroup analyses (e.g. types of DED including Sjögren's disease-associated dry eye) because data were sparse within comparisons.

In terms of interventions, this update evaluated multiple formulations of topical CsA, including nanoemulsions, nanomicellar solutions, and gel-based formulations with varying concentrations. While this reflects real-world therapeutic options, it also introduced substantial heterogeneity that limited the feasibility and interpretability of pooled analyses for several outcomes. In addition, the predominantly short follow-up period of three months in the included studies limited our ability to assess the long-term efficacy and safety of topical CsA.

Limitations of the review processes

We followed standard Cochrane methodology to minimize bias in conducting this systematic review. A Cochrane Eyes and Vision

Information Specialist used Cochrane highly sensitive search strategies to identify randomized trials in MEDLINE. We adhered to guidance in the Methodological Expectations of Cochrane Intervention Reviews (MECIR) standards when reporting the review [25].

Three authors (CSP, SCP, EKA) have financial conflicts of interest related to this review topic. One author (EKA) was the author of two included studies (Akpek 2023; Sheppard 2021), but she was not involved in the eligibility assessment, data extraction, risk of bias assessment, and GRADE assessments of those studies.

Agreements and disagreements with other studies or reviews

The findings of this review are broadly consistent with those of the 2019 version, which included 30 trials [24]. Overall, the certainty of evidence has increased from low in 2019 to low or moderate for some outcomes in this update. However, risk of bias and heterogeneity remain concerns. The direction of effect is also largely unchanged, indicating that improvements in DED signs and symptoms with CsA treatment remain variable and, in some cases, similar to vehicle or artificial tears. Of note, formulation-specific effects may vary. Tear production may improve with some CsA formulations (0.05% and 0.1%) and newer cationic emulsions. However, in general, the evidence for improvements in both tear production and tear film stability remains inconsistent and is limited to specific formulations with low-certainty evidence. In contrast, this review update provides more consistent evidence that CsA treatment is associated with increased treatment discontinuation due to adverse events (both water-free and cationic emulsion formulations), with moderate certainty for selected formulations.

Our review is broadly consistent with previous systematic reviews of CsA for dry eye disease. Across reviews, the most consistent finding is a small-to-moderate improvement in ocular surface staining compared with artificial tears or vehicle, although heterogeneity is substantial [151, 152, 153, 154, 155, 156].

Findings for other outcomes are more variable. Some reviews report improvements in tear film stability, tear production, and symptoms, while others show little or no difference [151, 153]. A recent review of water-free CsA found no significant improvement in corneal staining but suggested a clinically meaningful improvement in dryness scores [157]. Similarly, a meta-analysis in moderate-to-severe disease reported improvements across multiple outcomes but also identified increased adverse events and discontinuations [153]. Network meta-analyses likewise suggest potential benefits for symptoms and staining but highlight variability across formulations [158].

Tolerability findings are also inconsistent: while some reviews report higher adverse event and discontinuation rates with CsA [153, 158], others focusing on water-free formulations found comparable safety profiles between CsA and controls [157, 159].

Differences between other reviews and our findings likely reflect variation in inclusion criteria, disease severity, formulations, and analytic methods (e.g. change versus final scores), which may influence pooled estimates.

The clinical significance of improvements in symptoms and signs with CsA is generally small on average but may be meaningful for individual patients. Our findings indicate that clinically significant changes tend to occur primarily with specific formulations and are observed only in certain outcomes. However, the heterogeneity across trials and low- to moderate-certainty evidence limit confidence in these findings.

AUTHORS' CONCLUSIONS

Implications for practice

In this review update, low-certainty evidence suggested that topical cyclosporine A (CsA), in various formulations and concentrations, may improve both symptoms and signs of dry eye disease (DED) compared with vehicle or artificial tears alone. While these findings were generally consistent across outcomes such as corneal and conjunctival staining, results for subjective symptoms and tear production or stability were less consistent, with considerable statistical heterogeneity. These findings should be interpreted with caution due to variability in study design, outcome measures, follow-up durations, and overall methodological quality.

Although moderate-certainty evidence suggests that CsA probably slightly increases the risk of study discontinuation due to adverse events, the absolute rates of withdrawal were low.

The included studies represented a variety of DED etiologies, suggesting applicability to real-world clinical settings. However, due to the heterogeneity across interventions, outcome measures, and study populations, clinicians should tailor the choice of CsA formulation to individual patient profiles and clinical contexts.

Collectively, current evidence supports topical cyclosporine as a useful anti-inflammatory therapy for dry eye disease, primarily for improving objective ocular surface parameters. However, the magnitude of benefit is modest and the certainty of evidence ranges from low to very low for most clinically relevant outcomes.

Implications for research

Future research should aim to improve methodological quality and consistency across studies evaluating topical CsA for DED, especially by using a core outcome set for DED.

Future randomized controlled trials should use clearly defined participant populations, standardized outcome measures, and adequate follow-up to improve the comparability and applicability of findings. Comparative effectiveness studies are needed to evaluate different CsA formulations and concentrations, particularly across clinically relevant DED subtypes such as Sjögren's disease-associated DED and meibomian gland dysfunction. Longer-term studies are also required to assess whether treatment benefits are sustained and if prolonged CsA use is safe. Although beyond the scope of this review, inclusion of active comparators (e.g. lifitegrast or corticosteroids) in head-to-head trials would also provide valuable insights for treatment decision-making.

SUPPLEMENTARY MATERIALS

Supplementary materials are available with the online version of this article: [10.1002/14651858.CD010051.pub3](https://doi.org/10.1002/14651858.CD010051.pub3).

Supplementary material 1 Search strategies

Supplementary material 2 Characteristics of included studies

Supplementary material 3 Characteristics of excluded studies

Supplementary material 4 Characteristics of studies awaiting classification

Supplementary material 5 Characteristics of ongoing studies

Supplementary material 6 Analyses

Supplementary material 7 Data package

ADDITIONAL INFORMATION

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Editorial and peer-reviewer contributions

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The following people conducted the editorial process for this article:

- Sign-off Editor (final editorial decision): John Lawrenson, City St George's, University of London UK;
- Managing Editor (provided editorial guidance to authors, edited the article): Anupa Shah, Cochrane Central Editorial Service;
- Editorial Assistant (selected peer reviewers, conducted editorial policy checks, collated peer-reviewer comments and supported the editorial team): Andrew Savage, Cochrane Central Editorial Service;
- Copy Editor (copy editing and production): Faith Armitage, Cochrane Central Production Service;
- Peer-reviewers (provided comments and recommended an editorial decision): Dr Radhika Tandon, Professor of Ophthalmology, Chief, Dr. R. P. Centre for Ophthalmic Sciences, All India Institutes of Medical Sciences (AIIMS), New Delhi (clinical/content review); Tanuj Dada, AIIMS, New Delhi (clinical/content review); Antonella Lammoglia (patient and public review); Lindsay Robertson, Cochrane (methods review); Jo Platt, Central Editorial information Specialist (search review).

Contributions of authors

SP: screened search results for eligibility; assessed risk of bias; extracted data; entered and verified data; interpreted data; drafted the review; approved the final version for publication.

SS: assessed risk of bias; extracted data; entered and verified data; interpreted data; drafted the review; approved the final version for publication.

CCT: screened search results for eligibility; assessed risk of bias; extracted data; interpreted data; drafted the review; approved the final version for publication.

CN: screened search results for eligibility; assessed risk of bias; extracted data; interpreted data; drafted the review; approved the final version for publication.

CH: screened search results for eligibility; assessed risk of bias; extracted data; interpreted data; drafted the review; approved the final version for publication.

GH: interpreted data; drafted the review; approved the final version for publication.

CSP: supervised the review; interpreted data; drafted the review; approved the final version for publication.

SCP: supervised the review; interpreted data; drafted the review; approved the final version for publication.

EKA: supervised the review; interpreted data; drafted the review; approved the final version for publication.

TL: supervised the review; interpreted data; drafted the review; approved the final version for publication.

Cochrane Eyes and Vision: screened search results for eligibility; retrieved full-text reports; assessed risk of bias; extracted data; entered and verified data; conducted formal analyses; interpreted data; drafted the review; approved the final version for publication.

Declarations of interest

SP: none known

SS: none known

CCT: none known

CN: none known

CH: none known

GH: none known. GH is staff at Cochrane Eyes and Vision US Project, but was not involved in the editorial process for this review.

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What's new

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Registration and protocol

Protocol (2012): DOI: 10.1002/14651858.CD010051 [160]

Original review (2019): DOI: 10.1002/14651858.CD010051.pub2 [24]

Data, code and other materials

As part of the published Cochrane review, the following are made available for download for users of the Cochrane Library: full search strategies for each database ([Supplementary material 1](#)); full citations of each unique report and consensus risk of bias assessments for all studies included ([Supplementary material 2](#)), excluded at the full-text screen ([Supplementary material 3](#)), and awaiting classification ([Supplementary material 4](#)); analysis data ([Supplementary material 6](#)); and study data, including study information, study arms, and study results ([Supplementary material 7](#)). Appropriate permissions have been obtained for such use. Analyses and data management were conducted with Cochrane's authoring tool, Review Manager, using the inbuilt computation methods. Template data extraction forms from Covidence are available on reasonable request.

Date	Event	Description
17 July 2026	New citation required but conclusions have not changed	We screened the updated searches, and identified 32 eligible studies. Relevant sections have been updated.
17 July 2026	New search has been performed	Electronic searches were updated (studies not yet fully incorporated into review).

History

Protocol first published: Issue 9, 2012

Review first published: Issue 9, 2019

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ADDITIONAL TABLES

Table 1. Overview of syntheses and included studies

Intervention	Comparison	Study ID Study design	Condition(s) included	Number of randomized participants	Outcomes (method of synthesis)	Follow-up period(s)
Topical CsA 0.05% versus vehicle or artificial tears (AT) only						
0.05% CsA ophthalmic emulsion + AT (CMC 0.5%)	CMC 0.5%	Agarwal 2019 Parallel-group randomized controlled trial	People with dry eye	80 eyes of 40 participants	- Symptom (summary) - Conjunctival staining (summary) - Aqueous tear production (summary)	Six months

Topical cyclosporine A therapy for dry eye disease (Review)

Table 1. Overview of syntheses and included studies (Continued)

					• Tear film stability (summary)	
CsA 0.05% (Restasis) + AT (Refresh Tears)	AT (Refresh Tears)	Altiparmak 2010 Cluster-randomized controlled trial	People with thyroid orbitopathy	45 participants (73 eyes)	• Aqueous tear production (MA) • Tear film stability (MA) • Discontinuation due to AE (MA)	Six months
CsA 0.05% in Sophisen vehicle + AT (methylcellulose 0.5%)	Vehicle + AT (methylcellulose 0.5%)	Baiza-Durán 2010 Parallel-group randomized controlled trial (3 arms)	People with dry eye	183 participants (including another arm)	No outcomes of interest reported in this comparison	16, 21, 42, 70, and 98 days
CsA 0.05% + AT (sodium carboxymethylcellulose 0.5%)	AT (sodium carboxymethylcellulose)	Barreto 2009 Parallel-group randomized controlled trial	People with moderate dry eye secondary to HIV infection	20 participants	• Symptom (summary) • Conjunctival staining (summary) • Aqueous tear production (summary) • Tear film stability (summary)	Six months
CsA 0.05% + AT	Vehicle + AT	Brignole 2001 Parallel-group randomized controlled trial (3 arms)	People with moderate-to-severe KCS	169 participants (including another arm)	No outcomes of interest reported	Three and six months
CsA 0.05% + AT	Vehicle + AT	Chen 2010 Parallel-group randomized controlled trial	People with moderate-to-severe dry eye	233 participants	• Symptom (summary) • Corneal staining (MA) • Aqueous tear production (MA) • Tear film stability (MA) • AT usage (summary)	Two, four, and eight weeks
CsA 0.05% + AT	Vehicle + AT	Chen 2019 Parallel-group randomized controlled trial	People with moderate-to-severe dry eye	234 participants	• Symptom (summary) • Corneal staining (MA) • Aqueous tear production (MA) • Tear film stability (MA)	Seven, 28, 56, and 84 days
CsA 0.05% (Restasis)	Saline	Chung 2013 Intra-individual comparative randomized controlled trial	People with dry eye syndrome after cataract surgery	32 participants	• Symptom (summary) • Aqueous tear production (summary) • Tear film stability (summary)	Two weeks, one month, two and three months postoperatively

Table 1. Overview of syntheses and included studies (Continued)

CsA 0.05% (Restasis) + AT (Tears Naturale Free)	AT (Tears Naturale Free)	Demiryay 2011 Parallel-group randomized controlled trial	People with dysfunctional tear	42 participants	- Corneal staining (MA) - Conjunctival staining (MA) - Aqueous tear production (MA) - Tear film stability (MA) - Goblet cell density (MA)	Four months
- Preservative-free CsA 0.05% - Preservative-free CsA 0.05% + HA 0.1%	HA 0.1%	Gao 2023 Parallel-group randomized controlled trial (3 arms)	People with Primary Sjögren's Syndrome	60 participants	- Symptom (MA) - Corneal staining (MA) - Aqueous tear production (MA) - Tear film stability (MA)	Two, four, and 12 weeks
CsA 0.05% (Restasis) + AT (Tears Naturale Free)	Vehicle + AT (Tears Naturale Free)	Guzey 2009 Parallel-group randomized controlled trial	People with severe trachomatous dry eye	64 participants	- Symptom (MA) - Corneal staining (MA) - Conjunctival staining (MA) - Aqueous tear production (MA) - Tear film stability (MA) - Goblet cell density (MA)	One, three, and six months
CsA 0.05% + AT (0.1% sodium hyaluronate eye drops three times daily)	AT (0.1% sodium hyaluronate eye drops three times daily)	Hao 2024 Parallel-group randomized controlled trial	People with dry eye	80 participants	- Symptom (MA) - Corneal staining (MA) - Aqueous tear production (MA) - Tear film stability (MA)	One, two, and three months
CsA 0.05% (Restasis) + AT (Refresh Plus)	AT (Refresh Plus)	Kim 2009 Parallel-group randomized controlled trial (3 arm)	People with dry eye	150 participants (including another arm)	- Corneal staining (MA) - Aqueous tear production (MA) - Tear film stability (MA) - Goblet cell density (MA)	One, two, and three months
CsA 0.05% eye drop nanoemulsion + BAK-containing eye drops for glaucoma	AT (0.15% hyaluronic acid)+ BAK-containing eye drops for glaucoma	Kim 2023 Intra-individual comparative randomized controlled trial	People with dry eye and glaucoma	35 participants	- Symptom (MA) - Corneal staining (MA) - Aqueous tear production (MA) - Tear film stability (MA) - Discontinuation due to AE (MA)	Two and four months
0.05% CsA (Restasis) + 0.5% CMC (Refresh plus) 0.05% CsA (Restasis)	0.15% HA	Lee 2022 Parallel-group randomized controlled trial (3 arms)	People with severe dry eye	364 participants (including another arm)	- Symptom (MA) - Corneal staining (MA) - Tear film stability (MA) - Discontinuation due to AE (MA)	Four and 12 weeks

Table 1. Overview of syntheses and included studies (Continued)

+ 0.15% HA						
CsA 0.05% (Restasis) + AT (Visine Pure Tears)	Vehicle + AT (Visine Pure Tears)	Liew 2012 Parallel-group randomized controlled trial (7 arms)	People with dry eye	327 participants (including other arms)	- Symptom (summary) - Corneal staining (summary) - Conjunctival staining (summary) - Aqueous tear production (summary) - Tear film stability (summary) - AT usage (summary)	Eight weeks
CsA 0.05% + AT	Vehicle + AT	Ma 2015 Parallel-group randomized controlled trial	People with moderate-to-severe dry eye	40 participants	- Symptom (summary) - Conjunctival staining (summary) - Aqueous tear production (summary) - Tear film stability (summary)	One, four, eight, and 12 weeks, and two weeks after withdrawal
CsA 0.05% nanoemulsion (Cyporin N)	AT (0.15% HA)	Moon 2024 Randomized controlled trial	People with dry eye	35 participants	- Symptom (MA) - Corneal staining (MA) - Aqueous tear production (MA) - Tear film stability (MA) - Discontinuation due to AE (MA)	Four, eight, and 12 weeks
CsA 0.05% + AT	Vehicle	NCT04734197 Parallel-group randomized controlled trial (7 arms)	People with dry eye	94 participants	- Symptom (summary) - Aqueous tear production (MA) - Tear film stability (MA)	84 days
CsA 0.05% (Restasis)	0.1% sodium hyaluronate 0.15% sodium hyaluronate 0.3% sodium hyaluronate	Park 2017 Parallel-group randomized controlled trial (4 arms)	People with dry eye	176 participants	- Symptom (MA) - Corneal staining (MA) - Conjunctival staining (MA) - Aqueous tear production (MA) - Tear film stability (MA)	Six and 12 weeks
CsA 0.05% gel + AT	Vehicle + AT	Peng 2022 Parallel-group randomized controlled trial	People with dry eye	644 participants	- Corneal staining (MA) - Aqueous tear production (MA) - Tear film stability (MA) - Discontinuation due to AE (MA)	Two, six, and 12 weeks
CsA 0.05% (Restasis) + AT	AT	Prabhasawat 2012 Parallel-group randomized	People with symptomatic	70 participants	- Symptom (summary) - Corneal staining (summary) - Conjunctival staining (summary) - Tear film stability (summary)	One, two, and three months

Table 1. Overview of syntheses and included studies *(Continued)*

		controlled trial	Meibomian gland dysfunction			
CsA 0.05% (Restasis) + AT (Refresh Endura)	AT (Refresh Endura)	Rao 2010 Parallel-group randomized controlled trial	People with dry eye	74 participants	- Symptom (MA) - Corneal staining (MA) - Aqueous tear production (MA) - Tear film stability (MA)	Four, eight, and 12 months
CsA 0.05% nanoemulsion (Cy-porin N) + AT (0.15% HA) + warm compress	AT (0.15% HA) + warm compress	Rhim 2022 Parallel-group randomized controlled trial	Obstructive MGD	52 participants	- Symptom (MA) - Corneal and conjunctival staining (MA) - Aqueous tear production (MA) - Tear film stability (summary)	One and three months
CsA 0.05% (Restasis) + AT (Refresh)	AT (Refresh)	Salib 2006 Cluster-randomized controlled trial	People with dry eye syndrome undergoing laser in situ keratomileusis (LASIK)	21 participants	- Symptom (summary) - Aqueous tear production (summary)	One week, and one, three, six, and 12 months after surgery
CsA 0.05% + AT (Refresh)	Vehivle + AT (Refresh)	Sall 2000 Parallel-group randomized controlled trial (3 arms)	People with moderate-to-severe dry eye	877 participants (including another arm)	- Symptom (summary) - Corneal staining (summary) - Conjunctival staining (summary) - Aqueous tear production (summary) - AT usage (summary) - Discontinuation due to AE (MA)	One, three, four, and six months
- CsA 0.05% (Restasis) + AT (Refresh Tears) - CsA 0.05% (Restasis) + AT (Systane)	AT (Systane)	Sall 2006 Cluster-randomized controlled trial (3 arms)	People with aqueous deficient dry eye	61 participants	- Symptom (summary) - Corneal staining (summary) - Tear film stability (summary)	Seven, 14, 28, and 42 days and four and six months
CsA 0.05% + AT	AT	Schrell 2012 Randomized controlled trial	People with dry eye	62 participants	- Symptom (summary) - Corneal staining (summary) - Conjunctival staining (summary) - Aqueous tear production (summary) - Tear film stability (summary)	Three months

Table 1. Overview of syntheses and included studies (Continued)

CsA 0.05% + AT (Refresh)	Vehicle + AT (Refresh)	Stevenson 2000 Parallel-group randomized controlled trial (4 arms)	People with moderate-to-severe dry eye	162 participants (including other arms)	- Symptom (summary) - Corneal staining (summary) - Conjunctival staining (summary)	Four, eight, and 12 weeks and post-treatment two and four weeks
- CsA 0.05% nanomicrocella OTX-101 (CEQUA) - CsA 0.09% nanomicrocella OTX-101 (CEQUA)	Vehicle	Tauber 2018 Parallel-group randomized controlled trial (3 arms)	People with dry eye	454 participants	- Symptom (MA) - Corneal staining (MA) - Conjunctival staining (MA) - Aqueous tear production (MA) - Tear film stability (MA) - Discontinuation due to AE (MA)	14, 28, 42, 56, and 84 days
CsA 0.05% (Restasis)	AT (Refresh)	Willen 2008 Parallel-group randomized controlled trial	People with dry eye syndrome associated with contact lens wear	44 participants	- Symptom (MA) - Corneal staining (MA) - Aqueous tear production (MA) - Tear film stability (MA)	Three months
CsA 0.05% (Restasis)	Vehicle	Wirta 2019 Parallel-group randomized controlled trial (4 arms)	People with dry eye	207 participants (including other arms)	- Symptom (MA) - Corneal staining (MA)	Two, four, 12, and 16 weeks
CsA 0.05% + AT	AT	Wu 2009 Cluster-randomized controlled trial	People with dry eye	52 participants	- Aqueous tear production (summary) - Tear film stability (summary)	One, two, three, four, six, eight, and 12 weeks
CsA 0.05%	AT	Xu 2023 Parallel-group randomized controlled trial	People with dry eye	188 participants	- Symptom (MA) - Corneal staining (MA) - Aqueous tear production (MA) - Tear film stability (MA) - Discontinuation due to AE (MA)	Four and 12 weeks
CsA 0.05% versus other formulation of CsA						
CsA 0.1% in Sophisen vehicle + AT (methylcellulose 0.5%)	CsA 0.05% in Sophisen vehicle + AT (methylcellulose 0.5%)	Baiza-Durán 2010 Parallel-group randomized	People with dry eye	183 participants	- Symptom (summary) - Corneal staining (summary) - Aqueous tear production (summary)	16, 21, 42, 70, and 98 days

Table 1. Overview of syntheses and included studies *(Continued)*

	cellulose 0.5%)	controlled trial (3 arms)		(including another arm)		
CsA 0.1% + AT	CsA 0.05% + AT	Brignole 2001 Parallel-group randomized controlled trial (3 arms)	People with moder- ate-to-se- vere KCS	169 partici- pants (including another arm)	No outcomes of interest reported	Three and six months
CsA 0.09% + AT	CsA 0.05% + AT	Rajpoot 2022 Parallel-group randomized con- trolled trial	People with dry eye	467 partici- pants	- Symptom (summary) - Conjunctival staining (summa- ry) - Aqueous tear production (sum- mary) - Tear film stability (summary) - Discontinuation due to AE (MA)	One, two, and three months
CsA 0.1% + AT (Refresh)	CsA 0.05% + AT (Re- fresh)	Sall 2000 Parallel-group randomized controlled trial (3 arms)	People with moder- ate-to-se- vere dry eye	877 partici- pants	- Symptom (summary) - Corneal staining (summary) - Conjunctival staining (summa- ry) - Aqueous tear production (sum- mary) - Discontinuation due to AE (MA)	One, three, four, and six months
- CsA 0.1% + AT (Re- fresh) - CsA 0.2% + AT (Re- fresh) - CsA 0.4% + AT (Re- fresh)	CsA 0.05% + AT (Re- fresh)	Stevenson 2000 Parallel-group randomized controlled trial (4 arms)	People with moder- ate-to-se- vere dry eye	162 partici- pants (including another arm)	- Symptom (summary) - Corneal staining (summary) - Aqueous tear production (sum- mary) - Adverse events (MA)	One, three, four, and six months
0.1% cy- closporin A cationic emulsion (Ik- ervis) + AT	Restasis + AT	Jung 2023 Parallel-group randomized controlled trial	People with dry eye	56 partici- pants	- Symptom (MA) - Corneal staining (MA) - Conjunctival staining (MA) - Tear film stability (MA)	Four, eight, and 12 weeks
0.08% cy- closporine A na- noemulsion (TJO-087)	Restasis + AT (CMC 0.05%)	Eom 2023 Parallel-group randomized controlled trial	People with moder- ate-to-se- vere dry eye	178 partici- pants	- Symptom (MA) - Corneal staining (MA) - Conjunctival staining (MA) - Aqueous tear production (MA) - Tear film stability (MA) - Artificial tear usage	Eight, 16, 24, and 32 weeks
0.05% cy- closporin A nanoemul- sion + AT	Restasis + AT (sodium hyaluronate 0.1%)	Jo 2022 Parallel-group	People with MGD	90 partici- pants	- Symptom (summary) - Corneal staining (summary) - Conjunctival staining (summa- ry)	Four, eight, and 12 weeks

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Table 1. Overview of syntheses and included studies (Continued)

(sodium hyalur-onate 0.1%)		randomized controlled trial			• Aqueous tear production (summary) • Tear film stability (summary) • Adverse events (summary)	
0.05% cyclosporin A nanoemulsion + AT (sodium hyalur-onate 0.1%)	Restasis + AT (sodium hyaluronate 0.1%)	Kang 2020 Parallel-group randomized controlled trial	People with Sjögren syndrome associated with dry eye	80 participants	• Symptom (MA) • Corneal staining (MA) • Conjunctival staining (MA) • Aqueous tear production (MA) • Tear film stability (MA) • Goblet cell density (MA) • Discontinuation due to AE (summary)	Four and 12 weeks
0.05% CsA nanoemulsion formulation (Clac-ierTM)+AT (0.5% carboxymethyl-cellulose sodium; Hi-Eye Fresh eye drops)	Restasis + AT (0.5% carboxymethyl-cellulose sodium; Hi-Eye Fresh eye drops)	Kim 2017 Parallel-group randomized controlled trial	People with dry eye	101 participants	• Symptom (MA) • Corneal staining (MA) • Conjunctival staining (MA) • Aqueous tear production (MA) • Tear film stability (MA) • Discontinuation due to AE (MA)	Four, eight, and 12 weeks
0.05% CsA nanoemulsion (Cy-porin N) + AT (0.15% HA)	Restasis + AT (0.15% HA)	Park 2019 Parallel-group randomized controlled trial (3 arms, of them 2 arms were relevant to this review)	People with dry eye	150 participants (2/3 arms)	• Symptom (MA) • Corneal staining (MA) • Conjunctival staining (MA) • Aqueous tear production (MA) • Tear film stability (MA)	Four, eight, and 12 weeks
• 0.05% CsA gel once daily + AT (hypromellose) • 0.05% CsA gel twice daily + AT (hypromellose) • 0.1% CsA gel once daily + AT (hypromellose)	Restasis + AT (hypromellose)	Peng 2021 Parallel-group randomized controlled trial (4 arms)	People with dry eye	240 participants	• Symptom (MA) • Corneal staining (summary) • Aqueous tear production (summary) • Tear film stability (summary) • Adverse events (MA)	Four, six, and 12 weeks
• 0.05% CsA micellar nanoparticulate twice dai-	Restasis + AT (0.5% CMC, Refresh Tears)	Rao 2023 Parallel-group randomized	People with moderate to severe KCS	155 participants	• Symptom (MA) • Corneal staining (MA) • Conjunctival staining (MA) • Aqueous tear production (MA)	Four, eight, 12 and 24 weeks

Table 1. Overview of syntheses and included studies (Continued)

ly + AT (0.5% CMC, Re- fresh Tears)			controlled trial (3 arms)		- AT usage (summary) - Adverse events (summary)	
0.05% CsA micellar nanoparticulate twice daily for 4 weeks and once daily for 20 weeks + AT (0.5% CMC, Re-fresh Tears)						
- CsA0.01% + 3% trehalose +AT - CsA0.02% + 3% trehalose +AT - Placebo + AT	Restasis + AT	Shin 2021 Parallel-group randomized	People with dry eye	114 participants	- Symptom (summary) - Corneal staining (summary) - Conjunctival staining (summary) - Aqueous tear production (summary) - Tear film stability (summary) - Adverse events (summary)	Four, eight, and 12 weeks
CsA 0.1% versus vehicle						
CsA 0.1% cationic emulsion (Ikervis) + AT (saline solution)	Vehicle + AT (saline solution)	Baudouin 2017 Parallel-group randomized controlled trial	People with moderate-to-severe dry eye	495 participants	- Symptom (MA) - Corneal staining (MA) - Aqueous tear production (summary) - Tear film stability (MA) - Adverse events (MA)	Six months
CsA 0.1% unpreserved single-dose cationic emulsion (Ikervis) + AT (saline solution)	Vehicle + AT (saline solution)	Leonardi 2016 Parallel-group randomized controlled trial	People with severe dry eye disease	261 participants	- Symptom (MA) - Corneal staining (summary) - Conjunctival staining (MA) - Aqueous tear production (MA) - Tear film stability (MA) - Adverse events (MA)	One month, three, six, nine, and 12 months (nine and 12 months safety only)
CsA 0.1% water-free (CyclASol)	Vehicle only	Akpek 2023 Parallel-group randomized	People with dry eye disease	834 participants	- Symptom (MA) - Corneal staining (MA) - Conjunctival staining (MA) - Discontinuation due to AE (MA)	15 and 29 days

Table 1. Overview of syntheses and included studies (Continued)

controlled trial						
CsA 0.1% water-free (SHR8028)	Vehicle only	Peng 2024 Parallel-group randomized	People with dry eye disease	206 participants	- Symptom (MA) - Corneal staining (MA) - Conjunctival staining (MA) - Aqueous tear production (MA)	29 days
controlled trial						
CsA 0.1% water-free ophthalmic solution	Vehicle only	Sheppard 2021 Parallel-group randomized	People with dry eye disease	328 participants	- Symptom (MA) - Corneal staining (MA) - Conjunctival staining (summary) - Adverse events (MA)	Two, four, eight, and 12 weeks
controlled trial						
CsA 0.1% water-free ophthalmic solution	Vehicle only	Wirta 2019 Parallel-group randomized controlled trial (4 arms)	People with dry eye disease	207 participants	- Symptom (MA) - Corneal staining (MA) - Conjunctival staining (MA for CsA 0.05% and vehicle; summary for CsA 0.1%) - Discontinuation due to AE (MA)	Two, four, 12, and 16 weeks
controlled trial						
CsA 0.09% nanomicellar ophthalmic solution	Vehicle	Goldberg 2019 Parallel-group randomized	People with dry eye disease	745 participants	- Symptom (MA) - Corneal staining (MA) - Conjunctival staining (MA) - Discontinuation due to AE (MA)	28, 56, and 84 days
controlled trial						
- CsA 0.05% nanomicella OTX-101 (CEQUA) - CsA 0.09% nanomicella OTX-101 (CEQUA)	Vehicle	Tauber 2018 Parallel-group randomized controlled trial (3 arms)	People with dry eye	454 participants	- Symptom (MA) - Corneal staining (MA) - Conjunctival staining (MA) - Aqueous tear production (MA) - Tear film stability (MA) - Discontinuation due to AE (MA)	14, 28, 42, 56, and 84 days
controlled trial						
CsA 0.1% + AT	Vehicle + AT	Fan 2003 Non-parallel-group randomized controlled trial	People with Sjögren syndrome associated with KCS	50 participants	- Symptom (summary) - Conjunctival staining (summary) - Aqueous tear production (summary)	14, 30, 60, and 90 days
controlled trial						
CsA 0.1% + 0.5% CMC	0.5% CMC	Kudiyar 2018 Parallel-group randomized controlled trial	People with dry eye disease and who underwent cataract surgery	69 participants	Symptom (MA)	Four and eight weeks

Table 1. Overview of syntheses and included studies (Continued)

CsA 0.1% + AT (Refresh)	Vehicle + AT (Refresh)	Stevenson 2000 Parallel-group randomized controlled trial (5 arms)	People with moderate-to-severe dry eye	162 participants (including other arms)	- Symptom (summary) - Corneal staining (summary) - Aqueous tear production (summary) - Adverse events (MA)	Four, eight, and 12 weeks and post-treatment two and four weeks
CsA 0.1% versus CsA 0.1% + AT						
CsA 0.1%	CsA 0.1% + AT (HA)	Priani 2023 Randomized controlled trial (unit-of-analysis error)	People with moderate to severe dry eye	20 participants	- Symptom (MA) - Corneal staining (summary) - Aqueous tear production (MA) - Tear film stability (MA)	One month
CsA 1% versus vehicle or AT						
CsA 1% ointment + AT (Refresh)	Vehicle + AT (Refresh)	Laibovitz 1993 Cross-over-design randomized controlled trial	People with KCS	26 participants	- Symptom (Summary) - Conjunctival staining (summary) - Aqueous tear production (summary) - Tear film stability (summary) - Adverse events (summary)	Six weeks in each phase
CsA 1% + AT (0.3% HA)	AT (0.3% HA)	Shen 2020 Randomized controlled trial (unit-of-analysis error)	People with mixed types of dry eye	60 participants (120 eyes)	- Symptom (MA) - Corneal staining (MA) - Aqueous tear production (MA) - Tear film stability (MA)	One and three months
CsA 2% versus vehicle						
CsA 2% + AT (carboxy-polymethylene gel tear)	Vehicle + AT (carboxy-polymethylene gel tear)	Gündüz 1994 Parallel-group randomized controlled trial	People with secondary Sjögren syndrome	30 participants	- Conjunctival staining (MA) - Aqueous tear production (MA) - Tear film stability (MA) - Adverse events (MA)	Two months
CsA 2%	Placebo	Jain 2007 Cross-over-design randomized controlled trial	People with acquired primary lachrymal disease and Sjögren syndrome	30 participants	- Symptom (summary) - Conjunctival staining (summary) - Aqueous tear production (summary) - Tear film stability (summary)	Two months

AE: adverse event; **AT:** artificial tears; **BAK:** benzalkonium chloride; **CMC:** carboxy methylcellulose sodium; **CsA:** cyclosporine A; **HA:** hyaluronic acid; **KCS:** keratoconjunctivitis sicca; **MA:** meta-analysis; **MGD:** meibomian gland dysfunction; **TBUT:** tear film break-up time